

Remote-work and the restructuring of the firm : Evidence from collective agreements

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Abstract: We study how the formal adoption of remote work reorganizes the cost structure of the firm, using the universe of French collective agreements matched to administrative balance sheets over 2018–2025. A fine-tuned large language model classifies 400,000 agreements and extracts adoption timing and intensity for 20,000 firms in a staggered difference-in-differences design. Treated firms reduce their stock of office equipment and fittings by 22 percent at $t+5$ while sustaining production unchanged; in parallel, the wage bill falls through headcount reduction while average compensation per worker remains stable. Both adjustments unfold along the same contractual clock: France’s triennial baux 3-6-9 lease regime, and are absent among firms that formalized under COVID-19 emergency compulsion. Total factor productivity rises by 3.8 percent after five years, mechanically, because the same output is sustained with less capital. One to two days per week is the rationalization-maximizing intensity range; above three authorized days, coordination costs erode output and offset the real estate gain.

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1 Introduction

Remote work has become one of the defining organizational transformations of the past decade. Before the COVID-19 pandemic, telework remained marginal in most advanced economies: in France, only 9 percent of employees teleworked at least occasionally in 2019, and fewer than 4 percent did so on a regular basis (DARES, 2024). The pandemic triggered an abrupt and lasting shift. By 2024, more than one in five private-sector employees teleworks at least once a month, averaging 1.9 days per week (INSEE, 2025; Askenazy et al., 2025), and hybrid work has stabilized at around 44 percent of workers in teleworkable jobs across the European Union (Eurofound, 2024). This rapid diffusion raises a question that descriptive evidence alone cannot answer: through what mechanisms does the formal adoption of remote work reorganize the productive structure of the firm, and what determines whether that reorganization generates efficiency gains or coordination losses?

Answering this question requires three things that existing work has not simultaneously delivered. First, a direct measure of which firms adopted remote work and when, rather than a proxy based on occupational teleworkability. Second, a source of identifying variation that is independent of the firm’s efficiency trajectory. Third, a long enough horizon to observe the structural adjustments in capital, space, and workforce through which any cost-base rationalization actually materializes. Randomized experiments within single firms (Bloom et al., 2015, 2024) provide clean identification but are confined to specific organizations and short windows. Sector-specific analyses (Bergeaud et al., 2023a; Gibbs et al., 2023) extend scope but not horizon. Economy-wide studies (Tito, 2025) achieve breadth but must proxy for adoption, introducing attenuation bias and making it difficult to identify structural channels. This paper resolves all three limitations simultaneously.

Our setting is France over 2018–2025. Two institutional features make it uniquely suited to this question. The first is the legal architecture of collective bargaining. Since the September 2017 ordonnances Macron, remote work modalities are determined at the firm level through collective agreements that must be electronically filed on the Légifrance platform.

Because a valid agreement requires the signature of a representative trade union, the formalization date reflects a bilateral negotiation whose calendar is governed by union agendas and statutory bargaining rounds rather than by the firm’s efficiency outlook. We verify this directly: conditional on sector and year fixed effects, neither the level nor the recent trajectory of TFP predicts the timing of formalization among treated firms (Table 7). The second is the baux 3-6-9 commercial lease regime, under which leases run for a minimum of nine years with mandatory break clauses at three-year intervals. A firm wishing to reduce its office footprint after adopting remote work can only do so at these triennial renewal points. This contractual clock is entirely exogenous to the firm’s efficiency trajectory and generates sharp, predictable variation in the timing of real estate adjustment that we exploit directly.

We apply a fine-tuned large language model to approximately 400,000 collective agreements from 80,000 firms, constructing a direct, revealed measure of remote work adoption, its timing, and its authorized intensity for approximately 20,000 firms across all sectors and size classes. We then match these records to administrative balance sheet data and estimate dynamic treatment effects using a staggered difference-in-differences design with cohort-robust estimators (Sun and Abraham, 2021; Callaway and Sant’Anna, 2021). We use total factor productivity as a synthetic index of cost-base rationalization throughout: because the same output is sustained with fewer inputs, the ratio mechanically improves, and interpreting the movement as technological progress would be misleading. What we identify is an organizational restructuring event, not a shift in the production frontier.

The central finding is a coherent, two-margin restructuring of the firm’s cost base, governed by the contractual clock of the baux 3-6-9 regime. Output does not expand: treated firms sustain production unchanged while reducing their stock of office equipment and fittings by 22 percent at $t + 5$. In parallel, the wage bill falls through headcount reduction while average compensation per worker remains stable, ruling out wage compression as a mechanism. Both adjustments materialize along the same triennial timeline. We establish this restructuring channel through four complementary pieces of evidence. The timing of

the capital decline tracks the triennial lease structure precisely. Within the treated sample, the reduction is concentrated exclusively among firms whose lease falls due within one year of adoption and is absent for firms with longer residual terms, a within-treated contrast that is as good as random since the residual lease maturity at adoption is distributed approximately uniformly across the three possible positions in the triennial cycle, ruling out strategic timing of collective agreements in anticipation of an imminent break clause. The magnitude of the capital decline scales with local rents. And firms that formalized under COVID-19 emergency compulsion, without deliberate reorganization intent, exhibit no restructuring whatsoever, sharply falsifying organizational learning or worker motivation as alternative explanations. TFP rises by 3.8 percent at $t + 5$ as a mechanical consequence of this two-margin adjustment.

The restructuring channel is non-linear in intensity and heterogeneous across firm size, and we decompose these dimensions within the mechanisms analysis. A within-intensity input decomposition reveals the operative trade-off: capital declines monotonically with authorized days, from -4.0 log points at one day to -19.6 at five days, tracking the real estate channel at every intensity level. Output, by contrast, is stable at one and two days and falls significantly only at full remote work (-10.6 log points), consistent with the collaboration network evidence of [Yang et al. \(2022\)](#), who document that firm-wide remote work silos team networks and shifts communication toward asynchronous channels. The two-day arrangement is therefore not an arbitrary mass point in the distribution of agreements but the intensity at which real estate rationalization is maximized before coordination costs on output materialize. Size heterogeneity reveals three qualitatively distinct profiles. Large firms achieve output-neutral capital shedding, with TFP rising by $+5.2$ pp. SMEs reduce capital but without sufficient rental exposure to generate measurable net savings. Mid-sized firms present a profile in which output, wage bill, and intermediate inputs all contract simultaneously without TFP improvement, consistent with coordination costs dominating at the intensities they disproportionately adopt.

This paper connects to several strands of the literature. On firm reorganization and productivity, we differ from the existing literature in treating TFP not as an object of study but as a synthetic index of cost-base rationalization, turning the standard critique of difference-in-differences productivity estimates, namely that input declines mechanically inflate the residual, into the core finding. Our results validate the key prediction of [Sedláček and Shi \(2023\)](#) that capital cost savings are the dominant channel of remote work gains, while the mid-sized firm profile is consistent with their result that coordination frictions can dominate at sub-optimal intensities. On commercial real estate, our firm-level evidence provides the micro-level mechanism behind the delayed aggregate response of office values documented by [Gupta et al. \(2022\)](#) for the United States and [Bergeaud et al. \(2023b\)](#) for France: it is the triennial lease renewal cycle, operating firm by firm, that generates the discrete waves of demand reduction visible in aggregate office market data. On the labour channel, the [Li et al. \(2026\)](#) finding that the work-from-home wage premium reflects worker selection rather than a causal wage effect rationalizes our result that firm-level labour costs do not rise following adoption. Finally, our use of a fine-tuned large language model to classify multi-page legal documents builds on the text-as-data tradition surveyed by [Ash and Hansen \(2023\)](#), improving on the 77.5 percent accuracy achieved by [Askenazy et al. \(2025\)](#) on the same corpus to 98 percent sensitivity and 97 percent specificity via LoRA fine-tuning, a methodological improvement that is consequential for our binary adoption measure and the intensity extraction on which the heterogeneity analysis rests.

The remainder of the paper is organized as follows. Section 2 describes the institutional setting. Section 3 presents the data and LLM classification. Section 4 details the empirical strategy. Section 5 reports the TFP results. Section 6 decomposes the structural channels, including the capital and labour adjustments, the intensity trade-off, and heterogeneity by firm size and sector. Section 7 concludes.

2 Institutional Background

2.1 Collective Bargaining and Agreement Filing

French labor law requires firms with at least one union delegate to engage in periodic mandatory bargaining on compensation, working time, and work organization (Articles L.2242-1 to 2 of the Labor Code). Union delegates are appointed in firms with 50 or more employees, making this threshold the effective trigger for mandatory bargaining. When bargaining succeeds, the resulting agreement is legally binding and must be filed with the labor administration.

Two features of the September 2017 executive orders (*ordonnances Macron*) are central to our study. First, the reforms established the primacy of firm-level agreements over branch agreements in most domains, making the firm the default locus of negotiation for working time and work organization. This decentralization of bargaining to the firm level, documented as a Europe-wide trend by [Schoefer and Zulehner \(2024\)](#), created the institutional conditions under which telework modalities are now determined at the firm level in France rather than through sector-wide negotiation. Second, all firm-level agreements signed from September 1, 2017 onward must be deposited electronically on the *TéléAccords* platform and published on the *Légifrance* legal database. Each record includes the firm’s unique SIREN identifier, the signature date, and the full anonymized text. This digitized filing requirement creates the administrative paper trail that constitutes our primary data source and makes large-scale NLP feasible.

The timing of formal adoption within the treated sample reflects a bilateral negotiation rather than unilateral managerial discretion. This is important for identification: the union delegate’s objective function is partially decoupled from the firm’s short-run efficiency outlook, as union priorities center on employment protection, working conditions, and wage preservation rather than on input cost minimization. The calendar of bargaining rounds is governed by statutory deadlines (Articles L.2242-1 to 2) and union agenda constraints that

are independent of the firm’s position in the lease cycle. We verify this directly in Section 4: among treated firms, neither the level nor the trajectory of TFP predicts the timing of adoption. The 50-employee threshold triggering mandatory bargaining is set by law, providing an institutional basis for treating the negotiation calendar as partially exogenous to short-run efficiency shocks. The collective agreement channel is also the channel most likely to generate real estate reorganization: unlike a unilateral employer charter, a collective agreement formalizes a shared commitment to hybrid work that management can credibly use to justify footprint reduction in lease renegotiations.

2.2 The Legal Framework for Remote Work

Prior to 2017, implementing remote work required an individual amendment to each employee’s contract, creating substantial transaction costs. The 2017 ordonnances simplified this: remote work could henceforth be established through a collective agreement or, failing that, a unilateral employer charter, without individual contract amendments (Article L.1222-9 of the Labor Code). This reform made collective agreements the preferred legal vehicle for remote work, creating a direct link between our administrative data and actual adoption.

A firm-level remote work agreement is not a declaration of intent: it is a legally binding instrument specifying eligible positions, the number of authorized days, and management procedures. The post-pandemic ANI of November 2020, extended in April 2021, confirmed that specific modalities, including weekly intensity, are determined at the firm level.

Three channels currently exist for implementing remote work. The first is a collective agreement, filed on Légifrance. The second is a unilateral employer charter, which does not require union signature and is not publicly filed. The third is an informal individual arrangement between employer and employee. Our data capture only the first channel. We address this selection by restricting the control group to firms present in the ACCO-DTD database that engage in collective bargaining but never formalize remote work provisions

(Section 3).

2.3 The Commercial Lease Regime

The French commercial lease regime, known as *baux* 3-6-9, is central to our identification of the structural restructuring mechanism. Under this regime, commercial leases run for a minimum of nine years, with mandatory break clauses at three-year intervals. A tenant firm can exit or renegotiate its lease only at these triennial break points. This institutional feature creates a predictable and observable friction between the adoption of remote work and the effective rationalization of office space: a firm that formalizes telework at $t = 0$ typically faces its first opportunity to reduce its footprint between $t + 1$ and $t + 3$, depending on where in the triennial cycle its lease stands at adoption. This institutional clock is entirely exogenous to the firm’s efficiency trajectory, making it a clean source of variation for identifying the timing of the restructuring channel.

It is worth noting that this regime is specific to France. By contrast, commercial leases in the United States are typically negotiated individually with terms ranging from 3 to 15 years and no mandatory break clauses, giving American tenants more flexibility but also making the institutional clock less uniform. The French *baux* 3-6-9 regime therefore generates unusually sharp and predictable variation in the timing of real estate adjustment, which is what makes it useful for identification. The aggregate consequences of this lease structure for office market dynamics are documented by [Bergeaud et al. \(2023b\)](#), who show that the impact of telework diffusion on French office vacancies and prices unfolded gradually and differentially across departments in ways consistent with the rolling maturity of existing lease portfolios.

2.4 COVID-19 and the Diffusion of Remote Work Agreements

During the 2020 lockdowns, the government mandated remote work for all employees with compatible tasks, invoking the exceptional circumstances provision of Article L.1222-11 of

the Labor Code. This emergency regime did not require formal agreements. The wave of formal collective agreements peaked in late 2020 and early 2021 (Figure ??), reflecting deliberate organizational choices to institutionalize practices tested during the crisis. From mid-2021, the share of new remote work agreements stabilized at 4 to 5 percent of all monthly agreements, above the pre-pandemic baseline, indicating a lasting structural shift in how firms organize work.

Our staggered adoption design exploits variation in the timing of formal adoption, not in pandemic exposure. Pre-pandemic adopters (2018–2019) acted for reasons unrelated to COVID-19; crisis-era adopters (2020–2021) responded to the pandemic; later adopters (2022–2025) formalized arrangements during post-pandemic normalization. This heterogeneity underscores the importance of the cohort-robust estimators we employ, and it provides a natural falsification test of our restructuring mechanism, developed in Section 6: if the efficiency gain is driven by deliberate real estate reorganization, it should be absent among firms that formalized under regulatory compulsion rather than as part of a strategic reorganization of their cost structure.

3 Data

3.1 Collective Agreements

Since September 2017, French firms are required to file their collective agreements on the Légifrance platform.¹ Our database covers September 2017 through December 2025 and contains approximately 400,000 agreements from 80,000 firms. Each record includes metadata (SIREN, signature date, declared topics) and the full text of the agreement.

One limitation is that our data capture only remote work formalized through collective agreements, excluding unilateral charters and individual arrangements. We address this

¹This obligation was established by the ordonnances Macron of September 22, 2017, and primarily concerns firms with more than 50 employees that have a Comité Social et Économique (CSE).

by restricting the control group to firms present in the ACCO-DTD database that never formalize remote work through collective agreements, selecting controls comparable in their propensity to engage in firm-level bargaining. This sample restriction carries an identification advantage: among the three channels opened by the 2017 ordonnances, only the collective agreement requires the signature of a representative trade union, an actor whose objective function is partially decoupled from the firm’s short-run efficiency outlook. The timing of adoption therefore reflects the outcome of a bilateral negotiation rather than unilateral managerial discretion alone, attenuating the correlation between formalization dates and unobserved firm-specific efficiency trajectories.² We verify this claim directly in Section 4: among treated firms, neither the level nor the recent trajectory of TFP predicts the timing of adoption (Table 7).

Before turning to identification, we characterise the population of firms in the ACCO-DTD database along the dimensions most relevant to our analysis. Three features stand out. First, adoption rates rise sharply with firm size: 15.4 percent of SMEs formalise remote work through a collective agreement, against 36.4 percent of mid-sized firms (ETI) and 41.0 percent of large firms. Second, adoption is strongly concentrated in high-teleworkability sectors: rates reach 42.2 percent in real estate and 36.6 percent in ICT, against 8.3 percent in agriculture and 12.3 percent in construction (Table 4). Third, firms that adopted remote work entered collective bargaining earlier and more intensively than never-adopters. These imbalances imply that a naive comparison of adopters and never-adopters would confound the effect of remote work with pre-existing differences in scale, sector composition, and bargaining history. They directly motivate the inverse probability weighting procedure described in Section 4.

To quantify the predictability of non-adoption from structural observables, we estimate the following logistic model on the full unmatched population of 73,486 firms:

²The presence of a union delegate is itself institutionally determined by the 50-employee threshold triggering mandatory Comité Social et Économique representation. Since this threshold is set by law and not by firms, it provides an institutional basis for treating the timing of formal adoption as partially exogenous to short-run efficiency shocks.

$$\Pr(A_i = 0 \mid \mathbf{X}_i) = \Lambda(\gamma_0 + \gamma_1\tau_s + \gamma_2n_i + \gamma_3t_i^{\text{first}} + \gamma_4\mathbf{S}_i) \quad (1)$$

where $\Lambda(\cdot)$ is the logistic CDF, $\tau_s \in [0, 100]$ is the sectoral teleworkability index of [Sostero et al. \(2020\)](#), n_i is the number of collective agreements filed by firm i , t_i^{first} is the year of its first agreement, and $\mathbf{S}_i$ is a vector of firm-size indicators.

Table 1: Structural Determinants of Remote Work Non-Adoption (Unmatched Panel)

	Coef.	Std. Err.	Odds Ratio
Teleworkability index ($\hat{\gamma}_1$)	−0.0184***	(0.000)	0.982
Number of agreements ($\hat{\gamma}_2$)	−0.1900***	(0.002)	0.827
Year of first agreement ($\hat{\gamma}_3$)	0.0592***	(0.005)	1.061
Mid-sized firm (ETI)	−0.3234***	(0.025)	0.724
Large firm (GE)	−0.2962***	(0.034)	0.744
Observations		73,486	
Pseudo- R^2 (McFadden)		0.278	
Log-likelihood		−29,349	

Notes: Logit estimation of equation (1) on the unmatched panel. Dependent variable: indicator equal to one if firm i never files a remote work agreement. Teleworkability index from [Sostero et al. \(2020\)](#) ($\tau_s \in [0, 100]$). Reference size category: SME. Standard errors in parentheses. *** $p < 0.01$.

The results, reported in Table 1, confirm that non-adoption is highly predictable from structural observables: the McFadden pseudo- R^2 reaches 0.278. Three patterns are worth noting for their implications for identification. The teleworkability coefficient is negative and significant ($\hat{\gamma}_1 < 0$, OR = 0.982, $p < 0.001$), reflecting the strong sectoral sorting visible in Table 4. Bargaining intensity is also negatively associated with non-adoption ($\hat{\gamma}_2 < 0$, OR = 0.827, $p < 0.001$): the majority of never-adopters are firms with limited engagement in collective bargaining rather than firms that actively considered and rejected remote work arrangements. Finally, firm size is a strong predictor of adoption: large firms are substantially overrepresented among adopters relative to the unmatched population. Because size is correlated with the restructuring mechanism we identify in Section 6, larger firms face higher absolute rental costs and greater scope for footprint reduction at lease renewal, these raw differences underline the need to re-establish comparability before estimating causal ef-

fects. The inverse probability weighting procedure described in Section 4 addresses these imbalances directly: after reweighting, standardised mean differences on all pre-treatment observables fall below the conventional 0.10 threshold (Appendix F, Figure A5), and the gap between treated and control firms is substantially reduced (Table 6).

3.2 Balance Sheet Data

Annual financial statements come from tax filings deposited with commercial courts and disseminated by the INPI. Our extraction covers January 2017 through December 2025. Table 2 describes the accounting variables used to estimate the production function. Output is defined as revenue plus change in inventories. Capital is net tangible fixed assets. Labor is measured as wages and salaries plus social charges (the wage bill), which captures both worker qualification and hours worked in efficiency units. Intermediate inputs are raw materials purchases plus other purchases and external charges.

Table 2: Accounting Variables and Production Function Inputs

Variable	Definition	Accounting items
Y	Output	Revenue + change in inventories
K	Capital	Net tangible fixed assets
L	Labor	Wages and salaries + social charges
M	Intermediate inputs	Raw materials purchases + other purchases and external charges

We use the wage bill rather than headcount as our primary measure of labor input because it captures both worker qualification and hours worked (labor in efficiency units). The `prodestACF` implementation does not subtract the intermediate input contribution from the productivity residual; we recover $\hat{\alpha}_M$ via an auxiliary OLS regression and construct the corrected measure $\tilde{\omega}_{it} = \hat{\omega}_{it} - \hat{\alpha}_M m_{it}$. Since intermediate inputs decline post-adoption (Table A13), this correction works against finding an efficiency gain, providing a conservative test; estimates are robust to using the uncorrected measure (Appendix B). In robustness checks, we also re-estimate TFP using headcount as an alternative labor measure.

3.3 Local Rent Data

To test whether the capital rationalization channel is amplified in high-cost real estate markets, we append local rent data from the ANIL 2018 Carte des Loyers, which provides predicted median apartment rents at the commune level derived from approximately 9 million rental listings on SeLoger, leboncoin, and PAP. We match each firm’s headquarters commune via the SIRENE business register and aggregate to the department level using an observation-weighted median. The resulting variable, ℓ_d (euros per m² per month), ranges from 7.6 in Haute-Loire to 30.2 in Paris, with an inter-departmental standard deviation of 5.9. We construct a standardized interaction $\tilde{\ell}_d = (\ell_d - \bar{\ell})/\sigma_\ell$ and append it to the panel at 99.9 percent coverage.

We acknowledge that this variable measures residential rather than commercial rents. This is a limitation: ideally one would use commercial lease obligations directly. However, residential and commercial rents are highly correlated at the department level, as both reflect local land and congestion costs, as confirmed by the strong geographical co-movement documented in [Bergeaud et al. \(2023b\)](#). The interaction coefficient should therefore be interpreted as a lower bound on the true rent heterogeneity effect, and the conclusion section discusses this limitation explicitly.

3.4 LLM Classification of Agreements

The key challenge in exploiting collective agreements as a data source is their unstructured textual nature. We develop an automated classification pipeline based on a fine-tuned large language model.

Model architecture and fine-tuning. We use a 20-billion-parameter open-source LLM (Gpt-oss-20b), fine-tuned using Low-Rank Adaptation (LoRA; [Hu et al., 2022](#)). The training set consists of 300 manually annotated agreement-label pairs, chosen to represent the full range of sectoral contexts and agreement structures present in the ACCO-DTD database.

Fine-tuning hyperparameters are: LoRA rank $r = 16$, scaling coefficient $\alpha = 32$, learning rate $\eta = 2 \times 10^{-4}$, and 3 training epochs.

Classification performance. Table 3 reports the model’s performance on an independent test set of 100 manually annotated agreements not used during training. The fine-tuned model achieves 98 percent sensitivity and 97 percent specificity, substantially outperforming both keyword-based approaches and BERT models, which are constrained by a 512-token input limit incompatible with multi-page legal documents (see Appendix A for details, including an annotated example agreement). For reference, [Askenazy et al. \(2025\)](#) apply the Mixtral-8x7b-instruct model to the same Légifrance corpus without fine-tuning, achieving 77.5 percent accuracy on a 151-agreement validation sample. The LoRA fine-tuning procedure adopted here reduces this error rate by more than three-quarters, a material improvement given that classification errors translate directly into measurement error in the treatment variable and attenuation bias in all downstream estimates.

Table 3: Classification Performance of the LLM

	Sensitivity ("Remote work")	Specificity ("No remote work")
Base model (no fine-tuning)	95%	93%
Fine-tuned model (LoRA)	98%	97%
BERT (fine-tuned)	82%	78%
Keyword-based	71%	89%

Notes: Performance evaluated on 100 independently annotated agreements not used during training. Sensitivity: share of true remote work agreements correctly classified. Specificity: share of non-remote-work agreements correctly classified. BERT results are from a fine-tuned bert-base-multilingual-cased model truncated at 512 tokens. Keyword results use a dictionary of 47 French telework-related terms.

Extraction of remote work intensity. Beyond the binary classification, the model extracts the number of authorized remote work days per week using a structured JSON output format. We harmonize heterogeneous time units into days per week, assuming 47 working weeks per year. When agreements specify multiple regimes, we retain the modal regime. The

intensity variable is available for 38 percent of classified remote work agreements (approximately 6,900 firms); the distribution is strongly concentrated at two days per week (6,426 firms), with thinner right tails: 5,799 firms authorize one day, 1,824 three days, 488 four days, and 570 five days (Figure A1).

3.5 Panel Construction

We match collective agreements to balance sheet data using the SIREN identifier and enrich the panel with the SIRENE business register (legal category, industry code, firm size category). The initial sample contains 18,240 firms with at least one remote work agreement (23.4 percent of the 78,000 firms in the ACCO-DTD database). Merging with INPI data retains 54.9 percent of the target population; the final panel contains 7,731 treated firms and approximately 20,000 firms observed annually over 2018–2025, yielding approximately 105,000 firm-year observations.

The merge rate of 54.9 percent reflects three institutional features of French accounting law. Micro-enterprises (fewer than 10 employees, balance sheet below 450,000 euros) may request full confidentiality of their accounts under Article L.232-25 of the Commercial Code, rendering them invisible in the INPI public database. Second, small enterprises (fewer than 50 employees) may request partial confidentiality covering the income statement. Both mechanisms are orthogonal to the firms’ efficiency trajectories, consistent with the quasi-random attrition finding reported below.

We assess whether this attrition introduces selective bias. A logistic regression of attrition on all pre-merger observables yields a McFadden pseudo- R^2 of 0.011, indicating that attrition is quasi-random conditional on observables. The treated retention rate is 10.8 percentage points below that of controls, but the treated indicator is statistically insignificant in the logistic regression (coef. = -0.048 , $p = 0.585$), confirming that this gap reflects the greater propensity of larger treated firms to file consolidated rather than individual accounts. Under the most adverse assumption, that firms lost to attrition had TFP 20 percent above retained

firms, the five-year ATT remains at +1.6 pp, positive and economically significant. Full diagnostics are reported in Appendix E.

The never-treated group is defined as firms with no remote work agreement throughout the panel horizon, coded as $c_i = \infty$ in the Sun-Abraham framework. Firms whose first remote work agreement predates the panel start (2017 adopters, $N = 100$) are excluded from the treated sample as their pre-treatment dynamics are entirely unobserved; they are reassigned to the never-treated pool for the purpose of constructing the comparison group.³

3.6 Descriptive Statistics

The distribution of remote work agreements across the panel reflects the three phases visible in the raw filing data. Prior to the pandemic, remote work agreements represented 4 to 6 percent of all monthly collective agreements filed on Légifrance. A sharp spike exceeding 12 percent occurred in late 2020 as firms rushed to institutionalize practices tested during the lockdowns, before stabilizing at a post-pandemic plateau of 4 to 5 percent, a level that has remained stable through 2024. Using complementary survey and administrative sources, [Askenazy et al. \(2025\)](#) confirm that telework has entered a permanent regime in the French private sector, stabilizing at 23 percent of the private workforce with no recent signs of decline.

Table 4 shows the distribution of remote work agreements by firm size and sector. The adoption rate increases sharply with firm size: 15.4 percent of SMEs formalize remote work through a collective agreement, against 36.4 percent of mid-sized firms and 41.0 percent of large firms. The average authorized intensity decreases with size, at 2.24 days per week for SMEs against 1.86 for large firms, suggesting that larger firms regulate remote work more tightly once they adopt it. Adoption rates are highest in real estate (42.2 percent), ICT (36.6 percent), and finance (34.7 percent), and lowest in agriculture (8.3 percent) and construction

³Firms coded with $c_i = 0$ in the raw administrative data represent never-adopters and are recoded as $c_i = \infty$ accordingly. Sensitivity checks including 2017 pre-panel adopters as never-treated leave all results unchanged.

(12.3 percent). The median intensity across the full sample is 2 days per week, consistent with the finding of [Askenazy et al. \(2025\)](#) that the two-days-per-week hybrid model has become the dominant arrangement in French collective agreements. The concentration of adoption in large firms and high-teleworkability sectors is directly consistent with our heterogeneity results: these are precisely the firms with the greatest rental exposure and the broadest scope for real estate rationalization at lease renewal.

Table 5 compares treated and control firms before matching. Treated firms are substantially larger on average, with mean output of €78.8 million against €23.0 million for controls, and a correspondingly larger capital stock and wage bill. These imbalances confirm that a naive comparison of adopters and never-adopters would confound the effect of remote work with pre-existing differences in scale, sector composition, and bargaining history, directly motivating the inverse probability weighting procedure described in Section 4.

Table 4: Distribution of Remote Work Agreements by Firm Size and Sector

	Total	SME	Mid	Large	Agri.	Manuf.	Constr.	Trade	ICT	Finance	Real est.	Prof.	Public	Other
Remote work	18,240	6,988	6,858	3,832	88	4,188	741	3,258	1,636	1,300	479	4,050	2,725	1,178
No remote work	59,802	38,247	12,006	5,523	972	10,783	5,289	16,642	2,833	2,447	657	13,124	7,702	3,650
Total	78,042	45,235	18,864	9,355	1,060	14,971	6,030	19,900	4,469	3,747	1,136	17,174	10,427	4,828
Adoption rate	23.4%	15.4%	36.4%	41.0%	8.3%	28.0%	12.3%	16.4%	36.6%	34.7%	42.2%	23.6%	26.1%	24.4%
Days/week	2.05	2.24	1.89	1.86	1.59	1.75	1.60	1.82	2.65	1.87	1.62	1.95	1.69	1.77

Notes: Adoption rate = remote work agreements / total agreements. Days/week is the mean number of authorized remote work days per week among agreements reporting this information. Source: ACCO-DTD Légifrance, author’s calculations.

Table 5: Summary Statistics: Treated vs. Control Firms (Before Matching)

Variable	Treated			Controls		
	Mean	Median	SD	Mean	Median	SD
Output (€M)	78.8	20.6	298.8	23.0	6.6	123.7
Capital (€M)	11.4	0.9	152.5	3.4	0.3	56.3
Wage bill (€M)	13.7	5.3	60.9	4.4	1.8	21.9
Observations	25,836			163,494		

Notes: Statistics computed on pre-adoption years for treated firms and on the full period for controls. Output = revenue + change in inventories. Source: INPI, author’s calculations.

4 Empirical Strategy

TFP Estimation. Remote work may affect both labor and capital efficiency; we therefore estimate total factor productivity (TFP) from a gross-output Cobb-Douglas production function:

$$y_{it} = \alpha_K k_{it} + \alpha_L l_{it} + \alpha_M m_{it} + \omega_{it} + \varepsilon_{it} \quad (2)$$

where y_{it} denotes the log of output (revenue plus change in inventories), k_{it} capital, l_{it} labor, m_{it} intermediate inputs, ω_{it} efficiency observed by the firm but not the econometrician, and ε_{it} an unanticipated shock. This Cobb-Douglas specification is standard in the production function estimation literature ([Akerberg et al., 2015](#); [Levinsohn and Petrin, 2003](#)). OLS estimation of (2) suffers from simultaneity bias ([Marschak and Andrews, 1944](#)): a firm that anticipates higher efficiency will optimally choose higher input levels, generating a positive correlation between inputs and the efficiency residual. We adopt the [Akerberg et al. \(2015\)](#) estimator, which proxies unobserved efficiency with intermediate inputs and instruments labor with its lagged value. TFP is recovered as $\hat{\omega}_{it} = y_{it} - \hat{\alpha}_K k_{it} - \hat{\alpha}_L l_{it} - \hat{\alpha}_M m_{it}$, with the intermediate-input correction described in Section 3. As a robustness check, we also construct a calibrated gross-output residual with sector-specific input elasticities estimated on the control group following [Fiori et al. \(2025\)](#), yielding an aggregated ATT of +0.75 pp ($p < 0.01$) with a qualitatively identical dynamic profile; full results are reported in Appendix B.

Treatment Variable and Intent-to-Treat Interpretation. Our treatment variable is the signing of a collective agreement establishing remote work provisions, not actual employee take-up. Our estimates therefore carry an intent-to-treat (ITT) interpretation, which is the policy-relevant margin affected by labor law reforms (2017 ordonnances, 2020 ANI). This parameter is a conservative lower bound on the average treatment effect: DARES surveys indicate that 70 to 80 percent of eligible employees in firms with a remote work agreement actually use it ([DARES, 2023](#)). This compliance rate is further supported by [Askenazy et](#)

al. (2025), who show using matched individual and firm data from the French Labour Force Survey that the presence of a collective telework agreement is associated with a statistically significant increase in actual take-up probability, rising from 1.3 percentage points for one authorized day to over 3 percentage points for three or more days, even after controlling for firm-level fixed effects. This evidence implies that the gap between the ITT and the average treatment effect is moderate, and that our estimates should be read as a conservative lower bound on the true structural reorganization effect.

Inverse Probability Weighting. To address observable differences between treated and control firms (Table 5), we estimate the propensity score $e(X_i) = \Pr(D_i = 1 \mid X_i)$ via logit, separately by adoption cohort, using lagged covariates ($t - 1$): log output, log wage bill, log capital, and profit margin. Inverse probability weights are $w_i = D_i/e(X_i) + (1 - D_i)/(1 - e(X_i))$. After reweighting, all standardized mean differences fall below the conventional 0.1 threshold (Appendix F, Figure A5), and the gap between treated and control firms is substantially reduced (Table 6).

Table 6: Summary Statistics: Treated vs. Control Firms (After Matching)

Variable	Treated			Controls		
	Mean	Median	SD	Mean	Median	SD
Output (€M)	60.3	28.2	93.8	41.8	19.0	63.4
Capital (€M)	8.5	1.2	26.7	5.7	1.0	16.0
Wage bill (€M)	13.3	7.2	16.8	8.9	4.7	12.1
Observations	27,314			79,135		

Notes: Treated firms: all post-adoption firm-years. Controls: full period. Output = revenue + change in inventories. Source: INPI, author’s calculations.

4.1 Exogeneity of Adoption Timing

The IPW procedure addresses selection into treatment on observables. A remaining concern is that, among adopting firms, the timing of formalization may correlate with efficiency trajectories: firms experiencing efficiency declines might formalize remote work earlier in

search of cost reductions, or conversely, high-efficiency firms might adopt earlier because they can afford organizational experimentation. Either pattern would violate the parallel trends assumption.

To assess this concern, we restrict the sample to the 9,019 treated firms and retain only firm-year observations preceding their adoption date, forming a discrete-time risk set of 27,193 observations. For each firm-year, we define an indicator equal to one if the firm signs its remote work agreement the following year, and zero otherwise. We then estimate a sequence of logit models, progressively adding blocks of covariates, following the approach of [Malgouyres et al. \(2021\)](#). The goal is to assess whether pre-treatment efficiency levels or dynamics have explanatory power for the timing of adoption, conditional on sector and year fixed effects.

Table 7 reports the results. Sector $\times$ year fixed effects alone absorb 12.4 percent of the variance in adoption timing (column 2). Adding firm size raises the pseudo- R^2 by 0.47 percentage points (column 3), driven by the wage bill: larger firms formalize earlier, a selection pattern addressed by IPW. Adding the level and recent change in TFP raises the pseudo- R^2 by less than 0.01 percentage point relative to the fixed-effects-only model (column 4), with a joint Wald statistic of 0.21 ($p = 0.65$). Profitability is similarly uninformative ($F = 0.0004$, $p = 0.98$). In the full specification (column 6), the TFP block remains jointly insignificant while the size block is strongly significant ($F = 46.1$, $p < 0.001$). Conditional on sector $\times$ year fixed effects, the timing of formal remote work adoption is orthogonal to pre-existing efficiency trajectories.

4.2 Staggered Difference-in-Differences: Estimator Selection

Our identification strategy exploits temporal variation in remote work adoption across firms. The central assumption is parallel trends: absent treatment, the efficiency of treated firms would have evolved as that of control firms.

Since firms adopt remote work at different dates over 2018–2025, we face a staggered

Table 7: Explaining Adoption Timing Among Treated Firms

	(1) Covs	(2) S×Y FE	(3) +Size	(4) +TFP	(5) +Size+Marg	(6) +All
ln output	−0.005 (0.019)		0.003 (0.024)		0.003 (0.024)	0.004 (0.024)
ln wage bill	0.121*** (0.020)		0.146*** (0.025)		0.146*** (0.026)	0.145*** (0.026)
ln capital	−0.013 (0.008)		0.003 (0.010)		0.003 (0.010)	0.003 (0.010)
TFP				−0.015 (0.023)		
ΔTFP	−0.020 (0.057)			−0.026 (0.061)		−0.029 (0.063)
Margin	−0.176 (0.102)				−0.008 (0.114)	−0.002 (0.115)
Sector×Year FE	No	Yes	Yes	Yes	Yes	Yes
Pseudo R^2	0.003	0.124	0.128	0.124	0.128	0.128
Observations	27,193	26,158	26,158	26,158	26,158	26,158
Wald tests (column 6):						
TFP block (ΔTFP)				$F = 0.21, p = 0.649$		
Size block (output, wage bill, capital)				$F = 46.1, p < 0.001$		
Margin				$F = 0.0004, p = 0.984$		

Notes: Logit regressions on the sample of 9,019 treated firms observed before their adoption year (discrete-time hazard model). Dependent variable: indicator equal to one if the firm signs its remote work agreement the following year. TFP estimated via [Akerberg et al. \(2015\)](#) with wage bill as labor measure. ΔTFP is the year-on-year change. Margin is the ratio of operating surplus to output, winsorized at $[-1, 1]$. Column (1) includes all covariates without fixed effects. Column (2) includes sector × year fixed effects only. Columns (3) to (6) add covariate blocks sequentially. Standard errors: IID. Wald tests report joint significance of covariate blocks in the full specification (column 6).

adoption design. The standard two-way fixed effects (TWFE) estimator is biased in this setting when treatment effects are heterogeneous across cohorts ([Goodman-Bacon, 2021](#)): early-adopting cohorts may serve as implicit controls for later adopters, contaminating the estimates when treatment effect dynamics differ. We employ two complementary estimators that are robust to this issue.

For our main result, we use the [Sun and Abraham \(2021\)](#) estimator on the full unbalanced panel of 14,551 firms and 105,876 observations over 2018–2025. This estimator computes

clean weighted averages of cohort-specific ATTs, using never-treated firms reweighted by IPW as the comparison group. The dynamic treatment effect is estimated via the event-study specification:

$$\tilde{\omega}_{it} = \alpha_i + \delta_t + \sum_{k=-5}^5 \beta_k D_{it}^k + \varepsilon_{it} \quad (3)$$

where D_{it}^k indicates that firm i is k years from its adoption date, with $k = -1$ as the reference period.

For the mechanisms analysis, we use the [Callaway and Sant’Anna \(2021\)](#) doubly robust estimator on a balanced panel of 5,825 firms over 2017–2024 (46,600 observations), with not-yet-treated firms as the comparison group. This specification yields flat pre-trends for all inputs, a property that does not hold under Sun and Abraham on the full unbalanced panel, where output, labor, and intermediate inputs exhibit pre-trends at long horizons driven by sample composition rather than true violations of parallel trends. Appendix [B](#) confirms that the aggregate result is quantitatively similar under CS (ATT = 0.021, $p < 0.05$).

For the intensity heterogeneity analysis of Section [6.7](#), we complement the within-intensity TWFE sub-sample regressions with a pooled TWFE specification on the full panel that jointly estimates the effect of each intensity category using a common never-treated comparison group. This pooled specification, reported in Appendix [B.11](#), is the appropriate benchmark for comparing effects across intensity levels because it holds the control group constant. The sub-sample regressions in Table [9](#) are informative about the input decomposition within each intensity group but should not be used to rank intensities by TFP effect, as each sub-sample regression uses a different implicit comparison group.

Standard errors are clustered at the firm level throughout ([Bertrand et al., 2004](#)).

5 Results

5.1 Dynamic Treatment Effects on TFP

Figure 1 and Table A12 present the event-study estimates of the effect of remote work adoption on TFP, interpreted here as an index of input efficiency. The sample comprises 105,876 observations across 14,551 firms.

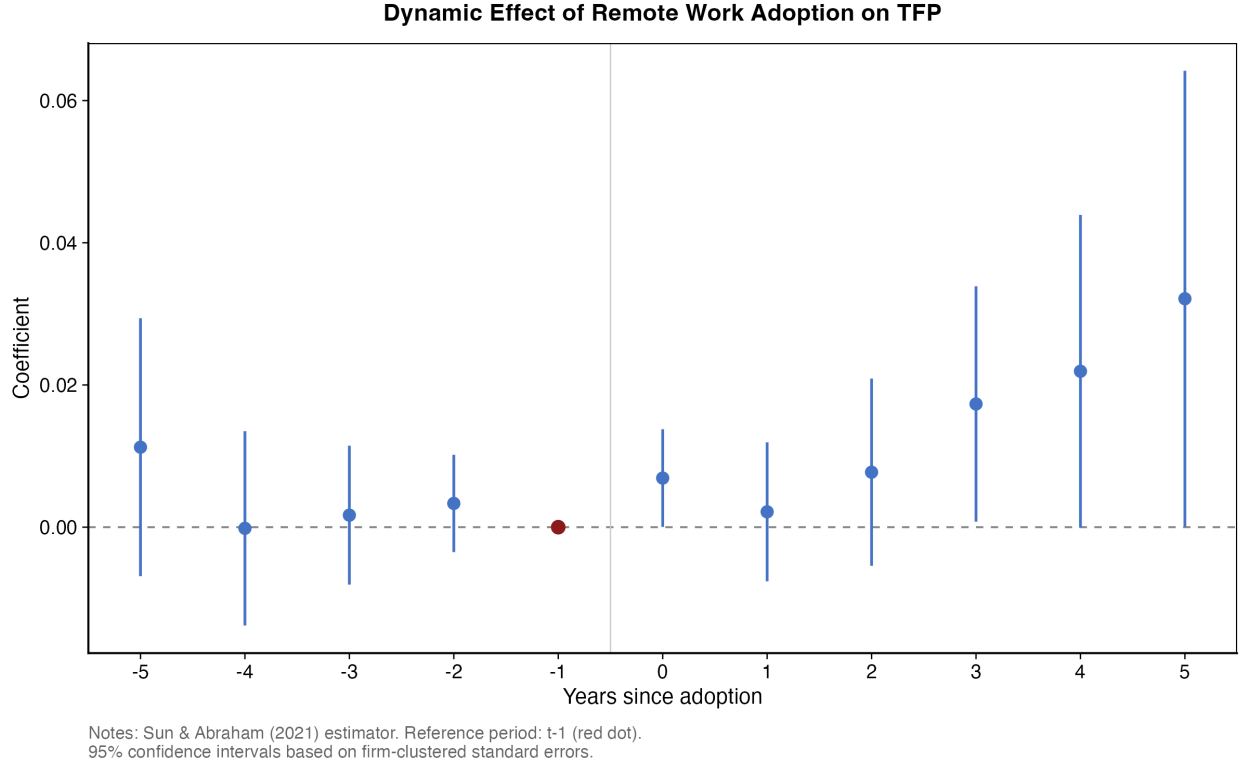


Figure 1: Dynamic Effect of Remote Work Adoption on Input Efficiency (TFP)

Notes: Event-study coefficients from equation (3), estimated using the Sun and Abraham (2021) estimator with IPW. TFP estimated via Akerberg et al. (2015) with the wage bill as the labor measure and corrected for the intermediate input contribution (see Section 3). Reference period: $t - 1$. Bars denote 95% confidence intervals based on firm-clustered standard errors. $N = 105,876$; 14,551 firms; Adj. $R^2 = 0.879$. Numerical coefficients reported in Table A12 (Appendix C).

The pre-treatment coefficients from $t - 4$ to $t - 2$ are small and statistically indistinguishable from zero (0.002, 0.003, and 0.004, respectively), supporting the parallel trends assumption. The coefficient at $t - 5$ is marginally significant (+1.4 pp, $p < 0.10$), consistent with the imprecision inherent in long pre-treatment windows in staggered designs. This ab-

sence of differential pre-trends is consistent with the finding (Table 7) that adoption timing is unrelated to pre-existing efficiency trajectories among treated firms.

The efficiency gain builds gradually from the third year of adoption. At $t = 0$, the effect is marginally significant (+0.7 pp, $p < 0.10$). Effects at $t + 1$ and $t + 2$ are modest and imprecisely estimated (+0.4 pp and +1.2 pp). From $t + 3$, the effect becomes statistically robust: +2.3 pp ($p < 0.01$), rising to +2.8 pp at $t + 4$ ($p < 0.05$) and +3.8 pp at $t + 5$ ($p < 0.05$). These are differential effects relative to control firms. The gradual build-up of the efficiency gain is the first piece of evidence pointing toward a structural rather than a behavioural mechanism: organizational learning or worker motivation effects would be expected to materialize quickly following adoption, whereas the observed trajectory follows precisely the contractual rhythm of triennial lease renewal, as we document in detail in Section 6.

Because our identification relies on formal collective agreements, it does not capture remote work implemented through unilateral employer charters or informal arrangements. To the extent that some firms in our control group may have adopted such practices, these estimates are subject to attenuation bias and should be interpreted as a conservative lower bound.

5.2 Robustness

The main finding is robust across estimators and specifications; full results are in Appendix B. The Callaway and Sant’Anna (2021) doubly robust estimator on the balanced panel yields a five-year point estimate of +4.1 pp with clean pre-trends and significant aggregated ATTs (+2.1 pp, $p < 0.05$; Table A1). The imputation estimator of Borusyak et al. (2024) yields a five-year point estimate of +2.5 pp and a monotonically increasing post-adoption profile (+0.5 pp at $t + 1$, +0.7 pp at $t + 2$, +1.7 pp at $t + 3$, +2.2 pp at $t + 4$), with pre-treatment coefficients uniformly small and centred on zero across the full $[-5, -1]$ window; the result is insensitive to the reweighting procedure and confirms that the main finding does not depend

on the choice of heterogeneity-robust estimator. Excluding firms formalizing telework during the COVID-19 period (2020–2021) leaves the five-year effect virtually unchanged at +3.9 pp ($p < 0.05$), with pre-trends remaining jointly insignificant (Figure A3). Replacing year fixed effects with sector $\times$ year fixed effects at the two-digit NAF level strengthens rather than attenuates the result (+5.3 pp at $t + 5$, $p < 0.01$; Table A4). A placebo treatment assigned three years before actual adoption yields a small, insignificant coefficient (-0.9 pp, $p = 0.13$; Table A5). Randomization inference over 1,000 permutations of adoption dates produces an exact p -value of 0.002 for the medium-run effect at $t + 3$ to $t + 5$ (Figure A2). Using headcount instead of the wage bill yields larger magnitudes (Table A3), consistent with the wage bill absorbing part of the efficiency gain through skill composition. Estimates are also insensitive to the intermediate-input correction to TFP (Table A6). Finally, replacing the ACF residual with a calibrated gross-output TFP measure with sector-specific elasticities estimated on the control group following [Fiori et al. \(2025\)](#) yields an aggregated ATT of +0.75 pp ($p < 0.01$) with a qualitatively identical dynamic profile; the result is insensitive to perturbations of ± 5 pp in any individual elasticity (Table A7).

Table A9 decomposes the aggregate ATT by adoption cohort. The cohort-specific ATT is largest and most precisely estimated for the 2019 cohort (+5.0 pp, $p < 0.01$), the only cohort with sufficient post-adoption years in the panel to reach $t + 5$ for the full treated sample. Cohorts adopting from 2020 onward show smaller and statistically imprecise effects. This pattern reflects a mechanical panel truncation constraint: our event-study identifies statistically significant efficiency gains only from $t + 3$ onward, and cohorts adopting in 2022, 2023, 2024, and 2025 reach at most $t + 3$, $t + 2$, $t + 1$, and $t = 0$ respectively by the end of the observation window in 2025. This truncation pattern, rather than heterogeneity in the structural effect of remote work adoption across waves, accounts for the declining cohort ATT profile. As a further check, Table A10 restricts the treated sample to pre-pandemic adopters only (cohorts 2018 and 2019, $N = 1,482$ firms), who have the longest post-adoption horizon in the panel and adopted for reasons entirely unrelated to the health crisis. The efficiency

gain is larger and more precisely estimated in this subsample, rising monotonically from +3.2 pp at $t = 0$ ($p < 0.05$) to +7.0 pp at $t + 5$ ($p < 0.01$), with flat pre-treatment dynamics throughout. This pattern is inconsistent with the interpretation that the aggregate result is driven by compositional effects from later, less precisely identified cohorts.

6 Mechanisms

The TFP movement documented in Section 5 is a synthetic signal: it tells us that the ratio of output to inputs has shifted, but not which inputs moved, in which direction, or through what institutional mechanism. This section decomposes that signal. We establish below that output is stable throughout the post-adoption window, that the dominant adjustment margin is capital, and that this capital adjustment is governed by the contractual clock of the baux 3-6-9 regime rather than by any diffuse organizational process. A secondary labour adjustment unfolds in parallel along the same timeline. The TFP movement is therefore not a measure of technological progress but an accounting summary of a two-margin cost-base reorganization: fewer desks and fewer workers producing the same output. We trace this reorganization in four steps: which asset categories decline and when; whether the adjustment is concentrated precisely among firms whose lease falls due shortly after adoption; whether it scales with local rents and task teleworkability; and whether it is absent when adoption was regulatory rather than deliberate.

6.1 Input Decomposition: Output Stability and Capital Shedding

Figure 2 presents event-study estimates for each production function input using the [Callaway and Sant’Anna \(2021\)](#) doubly robust estimator on the balanced panel of 5,825 firms over 2017–2024. The dominant result is the trajectory of capital (Column 3 of Table A13): following adoption, capital declines progressively and significantly relative to control firms, reaching 10.0 percent below controls at $t + 2$, 13.9 percent at $t + 3$, and 22.1 percent at

$t + 5$, against a perfectly flat pre-treatment baseline. Output is stable throughout, with post-treatment point estimates ranging from 1.3 to 2.7 percent, all insignificant. Labour and intermediate inputs show modest, imprecisely estimated declines from $t + 3$ onward, a pattern we decompose further in Section 6.6. The headline picture is therefore one of output-neutral capital shedding: treated firms sustain production while reducing their capital base. TFP rises mechanically as a result, because it measures the ratio of output to inputs and that ratio improves when inputs fall while output holds. The TFP gain is not the product of doing more; it is an accounting consequence of doing the same with less.

6.2 Decomposing the Capital Channel: Which Assets Decline, and When

Figure 3 presents the event-study estimates for these two sub-components. The cleanest signal comes from other tangible fixed assets (Panel A), the French GAAP category covering office furniture, fittings, and IT equipment tied to physical occupancy (Plan Comptable Général, 2019). This sub-component declines monotonically from $t + 1$, reaching 9.7 percent below controls at $t + 5$ ($p < 0.001$), with flat pre-treatment dynamics throughout. Constructions and land show no significant movement, which is informative: over 85 percent of service-sector firms above 50 employees are tenants carrying negligible owned real estate (INSEE, 2022). The relevant margin of restructuring runs through leases, not through balance sheet derecognition of owned property.

The rent proxy (Panel B) confirms the lease timing. Other external purchases decline significantly from $t + 3$ onward, reaching 2.5 percent below controls at $t + 5$. The delayed onset is an institutional fingerprint of the *baux* 3-6-9 regime: firms adopting at $t = 0$ face their first renewal opportunity between $t + 1$ and $t + 3$, producing precisely the observed timing.⁴

⁴The absence of any decline in constructions and land confirms that the mechanism operates through lease rationalization among tenant firms, not through disposal of owned property. The granular rental sub-component of other external purchases (code XQ) is unavailable at scale in the INPI filings; we rely on the

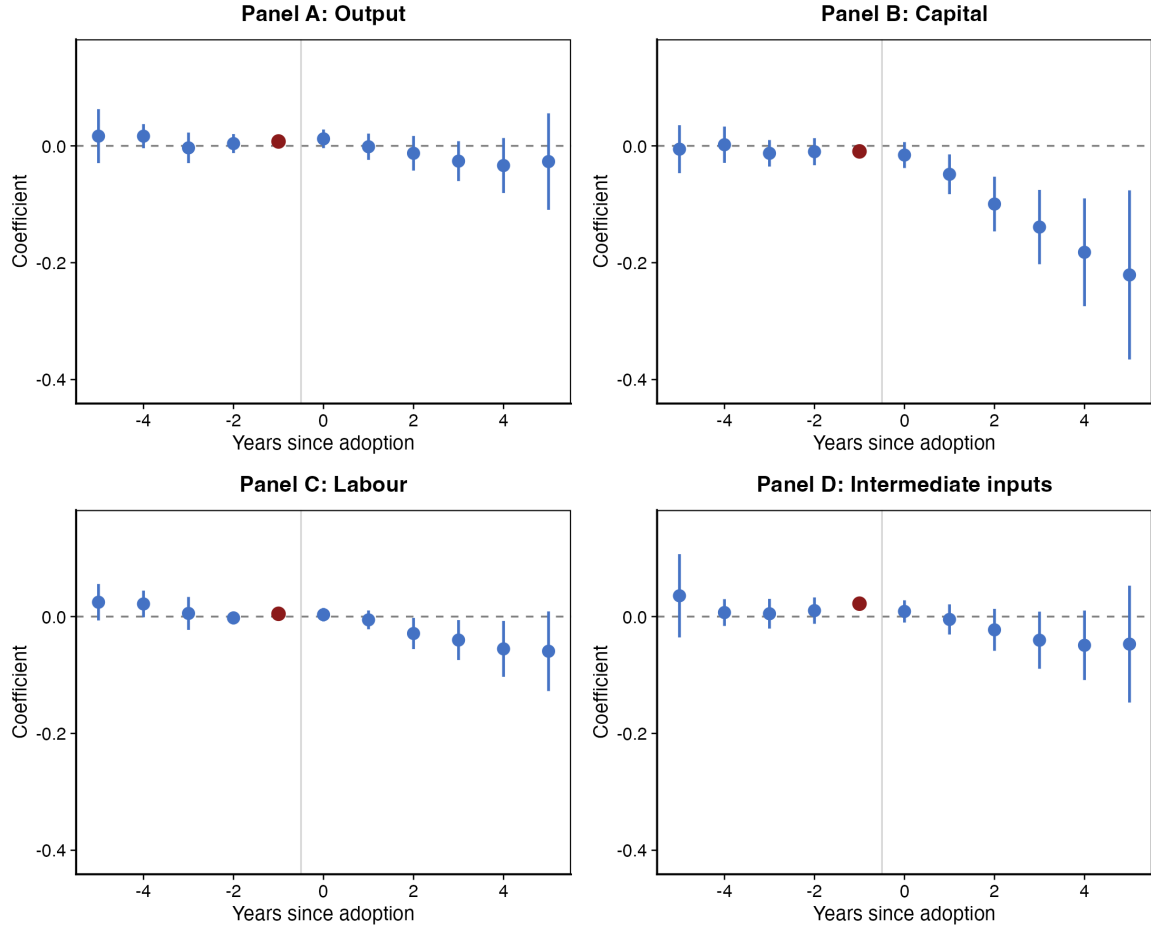


Figure 5: Event-Study Estimates for Production Function Inputs ([−5; +5] Window)
Notes: Callaway & Sant’Anna (2021) event-study coefficients with Doubly Robust estimation. Reference period: $t - 1$. Bars denote simultaneous 95% confidence

Figure 2: Dynamic Effects on Production Function Inputs: Callaway-Sant’Anna Estimates ([−5; +5] Window)

Notes: Callaway and Sant’Anna (2021) doubly robust estimator, not-yet-treated comparison group. Reference period: $t - 1$. Shaded bands: simultaneous 95% confidence intervals, firm-clustered bootstrap. Balanced panel, 5,825 firms, 2017–2024. TFP via Akerberg et al. (2015). Coefficients in Table A13 (Appendix C).

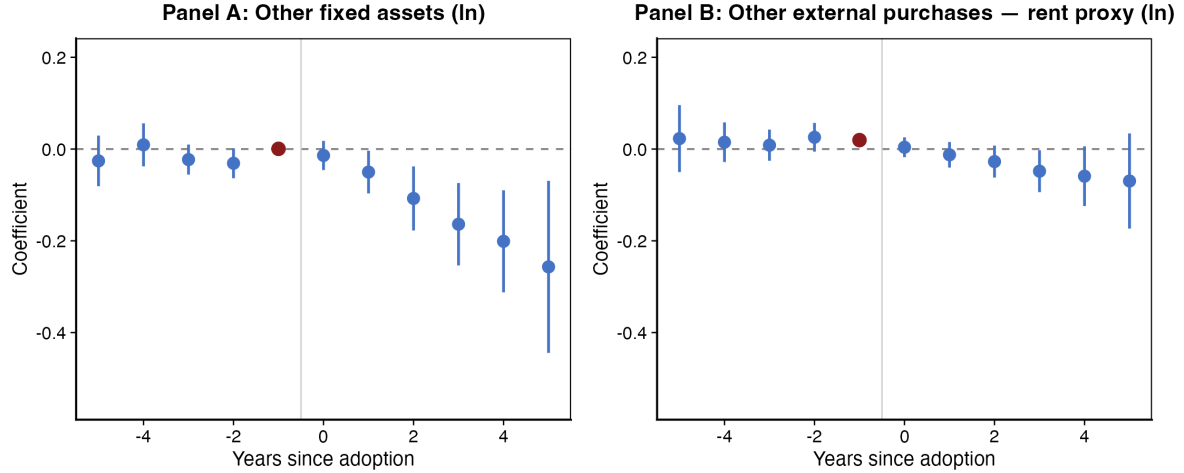


Figure 3: Decomposition of the Capital Channel

Notes: Callaway and Sant’Anna (2021) doubly robust estimator, not-yet-treated comparison group. Reference period: $t - 1$. Bars: simultaneous 95% confidence bands, firm-clustered bootstrap. Balanced panel, 5,825 firms, 2017–2024. Panel A: other tangible fixed assets (*Autres immobilisations corporelles*, French GAAP). Panel B: other external purchases as rental expenditure proxy (*Autres achats et charges externes*). * confidence band excludes 0.

6.3 Residual Lease Maturity at Adoption: A Within-Treated Test

The asset decomposition establishes what declines and when. A more demanding test asks whether the adjustment is driven specifically by the contractual window created by the *baux* 3-6-9 regime. We exploit variation that is internal to the treated sample.

Conditional randomness of the lease position at adoption. The residual maturity of the commercial lease at adoption depends on where in the triennial cycle the firm’s lease stands at the date of agreement signature, a feature determined by the historical lease signature date and orthogonal to any contemporaneous efficiency decision. A critical prerequisite is that firms do not time their remote work negotiations in anticipation of an imminent lease break. The data reject this hypothesis: the three residual maturity categories, zero, one, aggregate line as a conservative proxy, as competing sub-components have no mechanical reason to decline with remote work adoption.

and two years remaining, are each populated by approximately one third of treated firms. A chi-squared goodness-of-fit test against the discrete uniform distribution yields $\chi^2 = 1.48$ ($df = 2, p = 0.476$). This uniformity is further corroborated by a discrete-time hazard model estimated on treated firms whose lease was signed after 2008. Restricting attention to the pre-adoption window, we regress an indicator for adoption in year $t + 1$ on the residual lease maturity category, controlling for firm and year fixed effects. The coefficients on the zero- and one-year maturity categories are economically negligible and statistically indistinguishable from zero (-0.0015 , s.e. 0.0051 ; -0.0053 , s.e. 0.0051), and the distribution of adoption events across the three maturity categories is approximately uniform at one third each. Firms do not time their collective bargaining to coincide with an imminent lease break: the calendar of collective bargaining is determined by social negotiation dynamics independent of the lease cycle, so assignment to the short-maturity group is conditionally as good as random within the treated sample. Pre-adoption characteristics are balanced across the three groups, with standardized mean differences below 0.10 for all variables (Figure A6; Appendix G).

The within-treated contrast. Figure 4 presents event-study estimates of the effect of adoption on other tangible fixed assets separately for the two groups. The contrast is sharp: for firms locked into leases with more than one year remaining (Panel A), post-adoption coefficients are indistinguishable from zero throughout. For firms approaching a break clause at adoption (Panel B), office capital declines from $t + 1$ onward and deepens progressively through $t + 5$. Both groups adopt at $t = 0$ for the same institutional reasons and face the same organizational challenges; what differs is solely the timing of the contractual window. Capital adjustment follows this clock, not the adoption date itself.⁵

⁵Residual maturity is proxied by the administrative registration date of the firm’s headquarters in SIRENE, which typically coincides with the lease signature date. The sample is restricted to establishments registered after 2008 and to adoption cohorts 2018–2023, ensuring a symmetric pre-treatment window for all cohorts. The contrast is robust to replacing the Sun and Abraham (2021) estimator with the Callaway and Sant’Anna (2021) doubly robust estimator on the same trimmed sample: the group approaching an imminent break clause exhibits a significant and monotonically deepening capital decline from $t + 1$ onward (dynamic ATT = -0.47 , $p < 0.05$), while the group locked into longer leases shows coefficients indistinguishable from zero throughout the post-adoption window, with flat pre-trends in both groups.

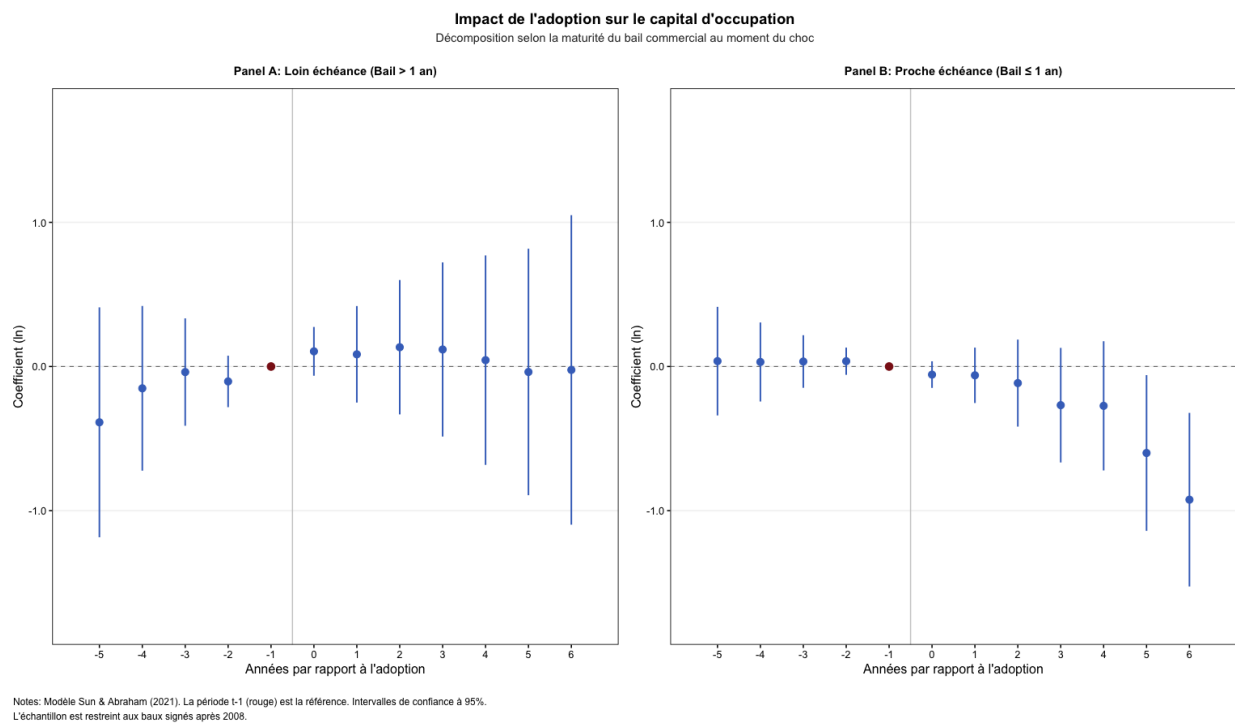


Figure 4: Impact of Adoption on Office Capital by Residual Lease Maturity at Adoption

Notes: Sun and Abraham (2021) estimator. Reference period $t - 1$ shown in red. 95% confidence intervals, firm-clustered standard errors. Sample restricted to establishments registered after 2011. Panel A: residual lease term > 1 year at adoption ($N \approx 2/3$ of treated firms). Panel B: residual lease term ≤ 1 year at adoption ($N \approx 1/3$ of treated firms).

6.4 Cross-Sectional Evidence: Local Rent and Teleworkability

If firms reduce their office footprint because remote work lowers the cost of doing so, the reduction should be larger where office space is more expensive. We estimate equation (4), interacting the treatment with the standardized local rent level $\tilde{\ell}_d$ and, separately, with the standardized sectoral teleworkability index $\tilde{\tau}_s$ of [Sostero et al. \(2020\)](#):

$$y_{it} = \alpha_i + \delta_t + \beta_1 D_{it} + \beta_2 (D_{it} \times \tilde{\ell}_d) + \varepsilon_{it} \quad (4)$$

where $\tilde{\ell}_d$ is absorbed by firm fixed effects and identified only through its interaction with D_{it} .

Table 8 reports the results. On log capital, both interactions are negative and significant: -0.035 per standard deviation of local rent ($p < 0.001$) and -0.026 per standard deviation of teleworkability ($p < 0.01$). On TFP, the rent interaction is marginally negative (-0.004 , $p = 0.064$), consistent with stronger wage renegotiation in high-cost urban markets absorbing part of the cost saving. Output is unaffected in both dimensions. The interaction coefficient is a lower bound on the total rental saving: treated firms in high-rent markets may additionally benefit from a price effect, as aggregate demand compression by co-located adopters depresses local office rents at renegotiation ([Gupta et al., 2022](#); [Bergeaud et al., 2023b](#)).⁶

6.5 Adoption Context: COVID Cohorts as a Consistency Check

A final piece of evidence comes from a dimension entirely independent of the time-series, within-treated, and cross-sectional tests above. The structural restructuring interpretation predicts that the efficiency gain should be absent among firms that formalized remote work under regulatory compulsion, since these firms did not make a deliberate decision to reorganize their productive footprint and therefore had no strategic reason to activate the lease renegotiation channel.

⁶The triple interaction $D_{it} \times \tilde{\ell}_d \times \tilde{\tau}_s$ is negative but imprecisely estimated ($p = 0.18$), suggesting the two channels operate in parallel rather than requiring both conditions simultaneously.

Table 8: Capital Rationalization, Local Rent, and Teleworkability

	ln capital (1)	TFP (2)	ln output (3)
D_{it} (treatment)	−0.014 (0.011)	0.010 (0.005)	0.012 (0.009)
$D_{it} \times \tilde{\ell}_d$	−0.035*** (0.006)	−0.004* (0.002)	−0.002 (0.005)
$D_{it} \times \tilde{\tau}_s$	−0.026** (0.009)		
Observations	105,800	105,800	105,800
Adj. R^2	0.939	0.884	0.952
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Notes: TWFE estimation of equation (4). $\tilde{\ell}_d$: standardized median apartment rent by département (ANIL 2018, z-score). $\tilde{\tau}_s$: standardized sectoral teleworkability index (Sostero et al., 2020), z-score. Both variables are absorbed by firm fixed effects; only their interactions with D_{it} are identified. The teleworkability interaction is estimated in a separate specification; TFP and output results for $\tilde{\tau}_s$ are omitted for brevity. Firm-clustered standard errors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 5 contrasts crisis-era adopters (2020–2021) with firms that formalized remote work outside the pandemic window (2017–2019 and 2022 onward). Crisis-era adopters exhibit no discernible efficiency gain across the post-adoption window, while non-pandemic cohorts display the positive, gradually building trajectory of Section 5. This pattern is directly consistent with the restructuring mechanism: firms that formalized under the emergency provisions of 2020–2021 did so in response to a regulatory mandate rather than as part of a deliberate reorganization of their cost structure, and therefore had little reason to activate the lease renegotiation channel. This result also rules out organizational learning or worker motivation as the primary drivers of the efficiency gain, since both would be expected to appear regardless of whether adoption was deliberate or mandated.

Taken together, the four pieces of evidence support a coherent and tightly identified restructuring channel: deliberate hybrid work adoption triggers a progressive reorganization of the firm’s office footprint, governed by triennial lease renewal cycles, concentrated among

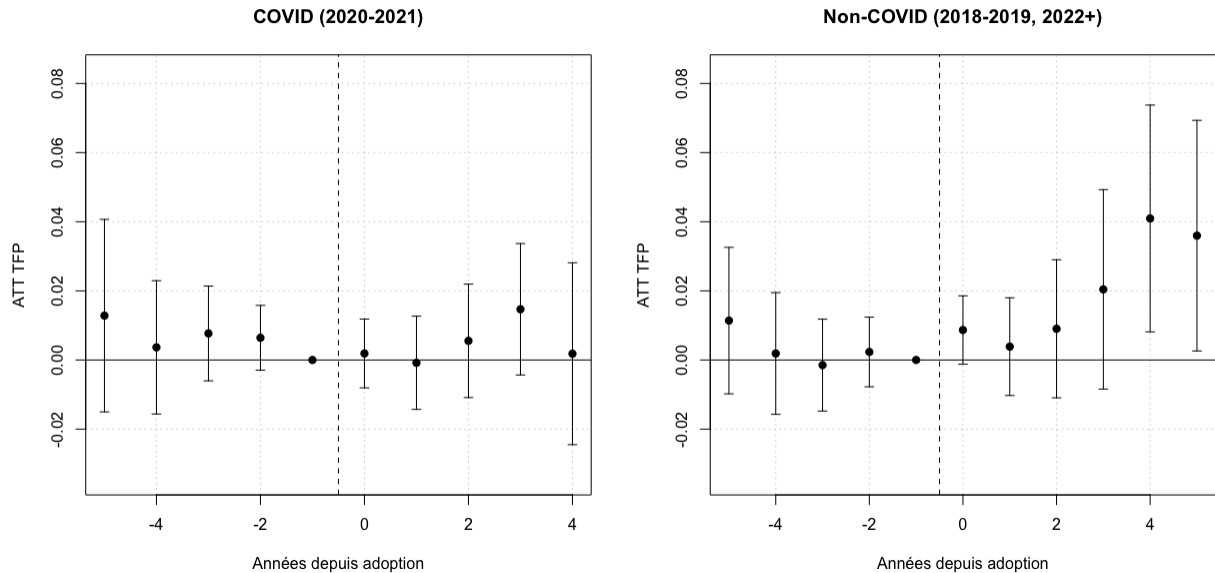


Figure 5: Event Studies by Adoption Context: COVID Cohorts vs. Non-COVID Cohorts

Notes: [Sun and Abraham \(2021\)](#) estimator with IPW. Left panel: firms adopting remote work in 2020–2021 (COVID-19 emergency period). Right panel: firms adopting outside the pandemic window (2017–2019 and 2022 onward). Reference period: $t - 1$. Bars denote 95% confidence intervals based on firm-clustered standard errors. Window: $[-5; +5]$.

firms whose lease falls due shortly after adoption, amplified where space is costly and tasks are teleworkable, and absent when adoption is purely regulatory. We now turn to the labour input, which reveals a parallel adjustment in workforce structure.

6.6 Decomposing the Labour Channel

The aggregate input decomposition of Figure 2 shows that labour costs decline modestly and imprecisely from $t + 3$ onward. This pattern is consistent with a gradual workforce adjustment unfolding in parallel with the real estate reorganization, but it leaves open a fundamental question: does the decline in labour costs reflect fewer workers, or lower average compensation per worker? These two interpretations have very different distributional implications. [Li et al. \(2026\)](#) document a work-from-home wage premium at the individual level in the United States, but show that this premium reflects worker selection rather than a causal wage effect: workers who sort into remote positions differ systematically in unob-

served productivity. This selection mechanism rationalizes why firm-level labour costs need not rise following adoption: if the premium is a selection premium, collective bargaining over telework need not raise average compensation, and any decline in the aggregate wage bill should be attributed to the extensive margin of employment rather than wage compression. To disentangle these margins, Figure 6 presents Callaway-Sant’Anna event-study estimates for three distinct labour measures: the log wage bill, the ratio of social contributions to the wage bill, and headcount.

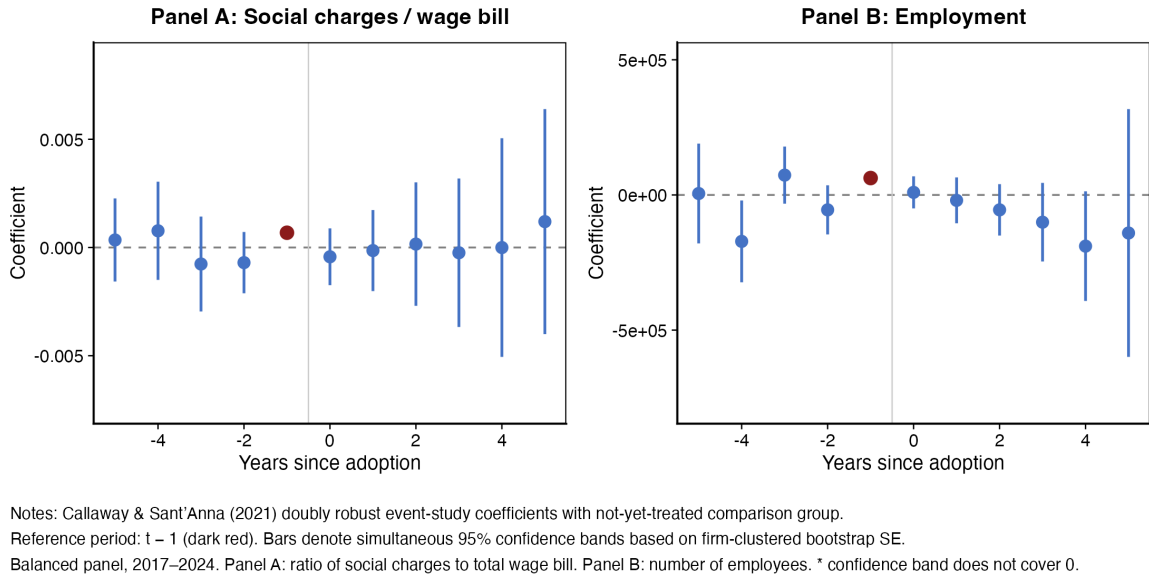


Figure 6: Decomposition of the Labour Channel: Callaway-Sant’Anna Estimates ($[-5; +5]$ Window)

Notes: Callaway and Sant’Anna (2021) doubly robust estimator, not-yet-treated comparison group. Reference period: $t - 1$. Shaded bands: simultaneous 95% confidence intervals, firm-clustered bootstrap. Balanced panel, 5,825 firms, 2017–2024. Panel A: log wage bill (ATT = -0.021 , s.e. 0.008 , significant at the 5% level). Panel B: social contributions as a share of the wage bill (ATT ≈ 0 , not significant). Panel C: headcount (ATT = $-56,355$, s.e. $38,222$, not significant). Full event-study coefficients reported in Appendix D.

Three results emerge. First, the log wage bill falls significantly and progressively relative to controls from $t + 3$ onward, reaching -0.055 log points by $t + 4$ ($p < 0.05$), with an overall ATT of -0.021 ($p < 0.05$), mirroring the timing of the capital decline precisely. Second, social contributions as a share of the wage bill are statistically indistinguishable from zero at every horizon (overall ATT = -0.0001 , s.e. 0.0009). Since social contribution rates in

France are strongly progressive and tied to individual compensation levels, a stable ratio rules out any decline in average compensation per worker. Third, the effect on headcount is negative and monotonically increasing in absolute magnitude from $t + 1$ onward, reaching an overall dynamic ATT of $-82,750$ ($p = 0.17$), but remains imprecisely estimated throughout, reflecting the elevated share of missing values for this variable in the *liasse fiscale* filings.

Taken together, these results identify headcount as the operative workforce adjustment margin: the wage bill falls while average compensation per worker is stable, implying that treated firms gradually right-size their workforce as they reorganize their office footprint at lease renewal. This workforce adjustment unfolds in the same time frame as the capital decline documented in Section 6.2, pointing to a coherent structural reorganization of the firm’s cost base, fewer desks and fewer workers producing the same output, rather than a piecemeal efficiency improvement in the technological sense. The distributional implication is that the cost savings from remote work adoption accrue to the firm through a leaner headcount, with the adjustment falling on the extensive margin of employment rather than on the intensive margin of wages.

6.7 Why Two Days? The Capital-Output Trade-off by Authorized Intensity

Table 9 documents how each production function input responds to adoption separately for each authorized intensity level, using sub-samples of never-treated firms and treated firms at the relevant intensity. These sub-sample regressions document input responses within each group; cross-intensity TFP comparisons require a common control group and are provided by the pooled specification in Appendix B.11.

Three results stand out. Capital declines monotonically with authorized intensity, from -4.0 log points at one day ($p < 0.10$) to -19.6 at five days ($p < 0.05$), significant at every level. This monotone gradient directly confirms that the real estate channel scales with intensity: more remote days release more desk space, generating proportionally larger

reductions at each lease renewal. Output is statistically zero at one and two days and falls significantly only at five days (-10.6 log points, $p < 0.10$), with three and four days showing a progressively negative but imprecise trend. The wage bill is unaffected throughout, ruling out any labour adjustment on the intensive margin.

The two margins therefore move in opposite directions at different thresholds. At one and two days, capital falls while output holds and the net effect on TFP is positive. Beyond three days, output begins to erode while capital continues to fall, and the coordination cost increasingly offsets the capital gain. This threshold is consistent with [Yang et al. \(2022\)](#): at low intensities, partial remote work reduces office occupancy costs without disrupting team networks enough to impair output; above two days, the siloing of collaboration networks and the shift to asynchronous communication begin to erode productive capacity. The pooled TWFE results confirm positive and significant TFP movements at one and two days ($+2.6$ pp and $+3.2$ pp), the range at which the rental saving registers without triggering output costs.

The monotone capital gradient provides additional evidence for the structural nature of the adjustment: a purely organizational mechanism would not generate capital scaling monotonically with authorized days while output holds at low intensities. The TFP optimum at two days is not a feature of any production technology; it is the intensity at which the rental saving is maximized before the output cost from network disruption materializes.

6.8 Input Decomposition by Firm Size

Table 10 reports Callaway-Sant’Anna aggregated ATTs separately for SMEs, mid-sized firms, and large firms. Three qualitatively distinct cost-adjustment profiles emerge.

Large firms: output-neutral capital shedding. Large firms present the canonical profile. Capital falls sharply (-12.2 log points, $p < 0.001$), accelerating from -9.2 log points at $t + 1$ to -21.7 at $t + 4$. Output is stable ($+3.3$ log points, $p = 0.10$), the wage bill is flat,

Table 9: Input Decomposition by Authorized Remote Work Intensity

	Authorized days per week				
	1 day (1)	2 days (2)	3 days (3)	4 days (4)	5 days (5)
<i>Panel A: TFP (ACF)</i>					
Remote work	+0.0250* (0.0152)	+0.0053 (0.0091)	−0.0029 (0.0174)	−0.0292 (0.0235)	−0.0735** (0.0344)
<i>Panel B: Output (ln)</i>					
Remote work	+0.0139 (0.0182)	+0.0183 (0.0124)	−0.0172 (0.0206)	−0.0360 (0.0293)	−0.1059* (0.0580)
<i>Panel C: Capital (ln)</i>					
Remote work	−0.0403* (0.0217)	−0.0549*** (0.0195)	−0.1637*** (0.0373)	−0.1953** (0.0661)	−0.1958** (0.0755)
<i>Panel D: Wage bill (ln)</i>					
Remote work	−0.0097 (0.0143)	+0.0173 (0.0115)	−0.0059 (0.0215)	+0.0040 (0.0292)	−0.0236 (0.0450)
Treated firms	1,324	1,807	505	139	169
Firm FE			Yes		
Year FE			Yes		

Notes: Each column is estimated from a separate TWFE regression on never-treated firms plus treated firms at the relevant intensity. Because the control group differs across columns, these coefficients should not be used to rank intensities by TFP effect; the pooled specification with a common control group is in Table A11 (Appendix B.11). Intensity categories: $[0.5, 1.5) = 1$ day, $[1.5, 2.5) = 2$ days, $[2.5, 3.5) = 3$ days, $[3.5, 4.5) = 4$ days, $\geq 4.5 = 5$ days. Results at 4 and 5 days are based on small treated subsamples. TFP via Akerberg et al. (2015), wage bill as labor measure. Firm-clustered standard errors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

and TFP rises by +4.9 pp ($p < 0.001$), reaching +10.8 pp at $t + 5$. The social contribution ratio is stable throughout. Large firms sustain production while shedding capital, consistent with the prediction of Sedláček and Shi (2023) that firms facing the highest fixed occupancy costs gain most from the reorganization channel.

SMEs: capital contraction without net savings. SMEs reduce capital significantly (-8.0 log points, $p < 0.01$); the rent interaction confirms the real estate channel is active (-0.023 , $p = 0.011$). Output declines marginally (-3.0 log points, $p = 0.05$) and TFP is flat. The mechanism operates, but below the scale threshold at which rental savings become large enough to offset coordination costs on output.

Mid-sized firms: the scissors profile. The aggregate ETI result, capital down (-6.6 log points, $p < 0.01$), output down (-2.6 log points, $p < 0.05$), wage bill down (-2.7 log points, $p < 0.05$), TFP flat, is a scissors profile in which all cost margins shrink but output does not hold. An intensity decomposition, reported in Table 11, qualifies this reading.⁷

Among ETI authorizing one to two days per week, the dominant adjustment is on capital, declining by -9.0 log points ($p < 0.05$) in the simple ATT and deepening monotonically to -25.4 log points at $t + 5$ ($p < 0.05$). Output, the wage bill, and TFP are statistically indistinguishable from zero, though their point estimates are consistently negative and build over time. The real estate channel is therefore active at moderate intensity, but the capital reduction does not translate into cost savings: at this scale, the absolute rental savings appear insufficient to offset even modest coordination costs on production. Among ETI authorizing three or more days, all four margins are statistically indistinguishable from zero; given the

⁷The not-yet-treated comparison group is unavailable for the ETI intensity decomposition: ETI adoption is sufficiently widespread that the not-yet-treated pool is empty for all nine cohorts within the balanced panel. The 992 never-treated ETI firms provide a comparison group with a treated-to-control ratio of 1.37 for the one-to-two-day subsample and 5.60 for the three-or-more-day subsample. The near-exhaustive adoption among ETI is itself informative: mid-sized firms formalized remote work earlier and more comprehensively than other size categories, consistent with stronger union pressure at this scale (Askenazy et al., 2025), and consistent with our argument that ETI adopt under negotiating pressure rather than as part of a deliberate reorganization strategy.

small treated subsample ($N = 153$), these results should be read as directionally suggestive rather than conclusive. Taken at face value, the pattern is consistent with coordination costs eroding both production and real estate rationalization simultaneously at high intensity.

The rent interaction sharpens the interpretation. On log capital, the ETI coefficient is the strongest of the three size categories (-0.047 , $p < 0.001$), confirming that ETI reduce their footprint most aggressively in expensive markets. On log output, the interaction is negligible (-0.001 , $p = 0.88$): production contraction is geographically uniform, consistent with a coordination cost that is independent of local real estate conditions.

Table 10: Callaway-Sant’Anna Estimates by Firm Size: Key Production Function Inputs

Outcome	SME		Mid-sized (ETI)		Large (GE)	
	ATT	SE	ATT	SE	ATT	SE
TFP (ACF)	−0.0053	(0.0108)	+0.0024	(0.0102)	+0.0489***	(0.0139)
Production (ln)	−0.0299**	(0.0151)	−0.0263**	(0.0121)	+0.0327	(0.0200)
Capital (ln)	−0.0799***	(0.0308)	−0.0656***	(0.0223)	−0.1224***	(0.0324)
Wage bill (ln)	−0.0220	(0.0137)	−0.0273**	(0.0120)	−0.0104	(0.0174)
TT × rent (capital)	−0.023**	(0.009)	−0.047***	(0.009)	−0.022	(0.012)
TT × rent (TFP)	+0.001	(0.004)	−0.008**	(0.003)	−0.005	(0.005)
TT × rent (output)	+0.005	(0.005)	−0.001	(0.005)	+0.003	(0.008)
Cylindrised firms	1,762		2,598		1,421	
Observations (CS)	14,096		20,784		11,368	
Observations (TWFE)	33,832		46,897		23,934	

Notes: Upper panel: [Callaway and Sant’Anna \(2021\)](#) doubly robust estimator, not-yet-treated comparison group, balanced panel 2017–2024. Lower panel: TWFE estimates of the interaction between treatment and standardized local rent ($\tilde{\ell}_d$, z-score), estimated separately by size category on the full unbalanced panel. Size categories: SME < 250 employees, ETI 250–4,999 employees, GE ≥ 5,000 employees. Firm-clustered standard errors in parentheses. · $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

6.9 Heterogeneity by Sector

Table [A14](#) (Appendix [C](#)) presents results by sector (Panel A) and by sector and size (Panels B to D). Three sectors exhibit large positive effects: finance and insurance (+14.1 pp, $p < 0.01$), ICT (+9.4 pp, $p < 0.01$), and professional services (+5.4 pp, $p < 0.01$). Two sec-

Table 11: ETI Input Decomposition by Authorized Intensity

	ETI, 1–2 days/week		ETI, 3+ days/week	
Outcome	ATT	SE	ATT	SE
<i>Panel A: Simple ATT</i>				
TFP (ACF)	+0.0033	(0.0151)	−0.0374	(0.0539)
Production (ln)	−0.0238	(0.0163)	−0.0521	(0.0346)
Capital (ln)	−0.0895**	(0.0291)	−0.0241	(0.0455)
Wage bill (ln)	−0.0242	(0.0179)	−0.0144	(0.0610)
<i>Panel B: Dynamic ATT (aggregated over [−5, +5])</i>				
TFP (ACF)	+0.0084	(0.0182)	−0.0661	(0.0698)
Production (ln)	−0.0345	(0.0230)	−0.0613	(0.0409)
Capital (ln)	−0.1166**	(0.0349)	−0.0190	(0.0560)
Wage bill (ln)	−0.0396	(0.0211)	+0.0063	(0.0815)
Treated firms	722		153	
Never-treated firms	992		992	
Control ratio	1.37		5.60	
Estimator	Callaway-Sant’Anna (2021), Doubly Robust			
Comparison group	Never Treated			
Panel	Balanced 2017–2024, ETI only			

Notes: Each column restricts the treated sample to ETI firms whose collective agreement authorizes the relevant number of days per week. Never-treated ETI firms serve as the comparison group in both specifications, as the not-yet-treated pool is empty for all nine ETI adoption cohorts within the balanced panel. The control ratio rows report the number of never-treated firms per treated firm. The capital decline among ETI at 1–2 days deepens monotonically to −25.4 log points at $t + 5$ ($p < 0.05$). Point estimates for ETI at 3+ days are uniformly negative but statistically imprecise ($N = 153$). Firm-clustered bootstrap standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

tors show significant negative effects: manufacturing (-2.4 pp, $p < 0.01$) and other services (-10.5 pp, $p < 0.01$). Figure 7 plots our estimated sectoral effects against the teleworkability index of [Sostero et al. \(2020\)](#). The correlation is positive and significant ($\rho = 0.79$, $p < 0.01$), consistent with a task-based interpretation ([Acemoglu and Autor, 2011](#)): sectors where the teleworkability of occupations is intrinsically higher are also those where the productive organization of work is most compatible with remote arrangements. This sectoral pattern directly mirrors the descriptive portrait of [Askenazy et al. \(2025\)](#), who document that the highest telework adoption rates in France are concentrated in the information-communication and financial services sectors, precisely the sectors where our causal estimates are largest. Crucially, these high-teleworkability sectors are also predominantly composed of large firms with expensive office footprints, which is why the lease restructuring channel is most operative precisely where teleworkability is highest: the two conditions, task compatibility and rental exposure, are jointly satisfied in the same part of the economy.

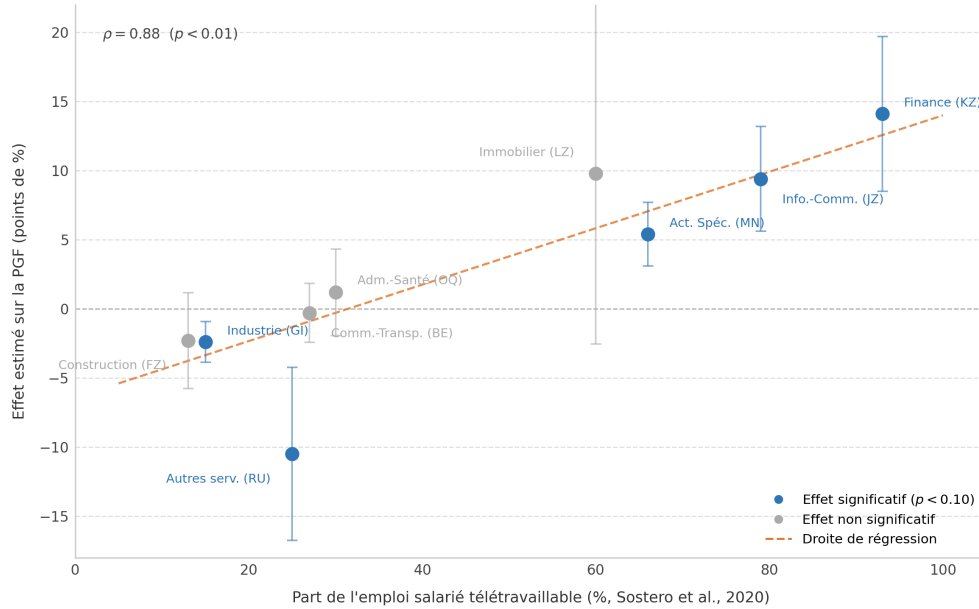


Figure 7: Estimated Remote Work Effects and Sectoral Teleworkability

Notes: Nine sectors (NA10 classification). Horizontal axis: share of teleworkable employment from [Sostero et al. \(2020\)](#), JRC WP 2020-05 (EU-27, LFS 2018–2019). Vertical axis: DiD coefficient from Panel A of Table A14; bars denote 90% confidence intervals. Regression line weighted by sectoral employment (INSEE, 2022). Gray points: not significant at the 10% level. Correlation coefficient $\rho = 0.79$ ($p < 0.01$). Sources: [Sostero et al. \(2020\)](#); author's calculations.

7 Conclusion

This paper studies how the formal adoption of remote work reorganizes the cost structure of the firm, using the universe of firm-level collective agreements matched to administrative balance sheet data over 2018–2025 in France. We decompose the structural adjustments in capital, space, and workforce through which this reorganization materializes, pin down the institutional mechanism that governs its timing, and trace how the gains and costs are distributed across firm sizes and intensities. Total factor productivity serves throughout as a synthetic index of cost-base rationalization: because the same output is sustained with fewer inputs, the ratio mechanically improves, and interpreting its movement as a measure of technological progress would be misleading. What the data show is an organizational restructuring event, not a shift in the production frontier.

Five results stand out. First, we document a coherent, two-margin restructuring of the firm’s cost base, governed by the contractual clock of the baux 3-6-9 lease regime. Treated firms reduce their stock of office equipment and fittings by 22 percent at $t+5$ while sustaining output unchanged; in parallel, the wage bill falls through headcount reduction while average compensation per worker remains stable. Both adjustments materialize along the same triennial timeline. The gradual buildup is not a signature of slow organizational learning but of the contractual constraints of commercial lease renewal. Four pieces of evidence support this channel: the timing of the capital decline tracks the triennial lease structure; within the treated sample, the reduction is concentrated among firms whose lease falls due within one year of adoption and is absent for firms with longer residual terms; the decline is larger in high-rent markets (-3.5 log points per standard deviation of local rent, $p < 0.001$); and firms that formalized under regulatory compulsion exhibit no restructuring. These micro-level results are directly consistent with the aggregate office market dynamics documented by [Bergeaud et al. \(2023b\)](#) for France and by [Davis et al. \(2024\)](#) for the United States: the baux 3-6-9 clock, operating firm by firm at lease expiration, is the micro-level engine generating the aggregate demand reductions for office space they document.

Second, a within-intensity input decomposition reveals why two authorized days per week is the rationalization-maximizing arrangement. Capital declines monotonically with intensity, from -4.0 log points at one day to -19.6 log points at five days, tracking the real estate channel at every level. Output, by contrast, is stable at one and two days and falls significantly only at full remote work (-10.6 log points, $p < 0.10$), while the wage bill is unaffected throughout. The two-day arrangement is therefore the intensity at which the rental saving is maximized before the output cost from network disruption materializes, consistent with the collaboration network evidence of [Yang et al. \(2022\)](#). The TFP movement at each intensity level is a direct consequence of this capital-output trade-off: it is positive where capital falls while output holds, and negative where output erosion dominates.

Third, size heterogeneity reveals three qualitatively distinct cost-adjustment profiles. Large firms achieve the canonical output-neutral capital shedding: capital falls by -12.2 log points, output is stable, and TFP rises by $+4.9$ pp, consistent with the prediction of [Sedláček and Shi \(2023\)](#) that firms with high fixed occupancy costs gain most from the reorganization channel. SMEs reduce capital but without sufficient rental exposure to generate measurable net savings. Mid-sized firms (ETI) present the most policy-relevant profile: they contract simultaneously on output, wage bill, and intermediate inputs without any TFP improvement, a scissors profile in which all cost margins shrink but output does not hold. The social contribution ratio is stable across all size categories, confirming that labour adjustments operate on the extensive margin of employment throughout.

These results carry concrete implications. For firms, moderate hybrid arrangements formalized through collective bargaining appear to be a lever for real estate rationalization and gradual workforce right-sizing, but the gains are concentrated at two days per week and among large firms with sufficient rental exposure. The transition is gated by the triennial lease cycle, and firms should anticipate a multi-year horizon before the cost savings fully materialize. For policymakers, the ETI scissors profile warrants specific attention: mid-sized firms appear to be absorbing the coordination costs of hybrid work without accessing

the restructuring gains, a distributional outcome that current policy instruments may not adequately address. The mechanism also points to a structural consideration for commercial real estate markets: as firms rationalize their office footprints at lease renewal, aggregate demand for commercial space adjusts in discrete waves rather than continuously, generating the vacancy and price dynamics documented by [Bergeaud et al. \(2023b\)](#) and [Gupta et al. \(2022\)](#). These waves will intensify as the large cohort of leases signed before 2020 reaches their triennial break points over the coming years.

Several limitations warrant mention. Our intent-to-treat design captures the effect of formal adoption rather than actual usage, though high compliance rates suggest the two are closely related. Our data do not capture remote work implemented through unilateral charters or informal arrangements, which may be more prevalent among SMEs and may partly explain their weaker response. The local rent proxy measures residential rather than commercial rents; direct data on office lease obligations would allow a sharper test of the channel and a more precise scaling of the rental exposure mechanism by firm size. The seven-year window may not capture the full long-run equilibrium effects of remote work on firm organization, particularly as office lease markets reprice in response to aggregate demand shifts. Finally, cohort-specific estimates for post-2021 adopters are estimated over short post-adoption horizons and will become more precisely identified as the panel extends beyond 2025.

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A LLM Classification

A.1 Why a Large Language Model?

Three approaches were considered for classifying collective agreements.

Keyword-based methods using regular expressions and keyword searches (e.g., matching the term *télétravail*) are transparent and reproducible but insufficient for our purposes. The term *télétravail* may appear in an agreement merely to announce future negotiations or to reference a pre-existing branch-level agreement, without the firm itself establishing remote work provisions. Conversely, the formulations used to institute remote work vary widely across agreements (*travail à distance*, *modalités de travail hybride*, *organisation du travail en mode distanciel*, etc.), making exhaustive pattern-matching impractical.

Bidirectional Encoder Representations from Transformers (BERT; [Devlin et al., 2019](#)) offer more sophisticated semantic understanding, but their architecture imposes a hard limit of 512 tokens (approximately one page of text). Collective agreements frequently span several dozen pages. In our tests, a fine-tuned BERT model achieved 82 percent sensitivity and 78 percent specificity on the same test set, considerably below the LLM’s performance (Table 3).

Large Language Models process long documents natively (up to 128,000 tokens for the model we use) and perform complex semantic reasoning through natural-language prompting. These capabilities make them well suited to our task, which requires understanding the overall intent of a multi-page legal document rather than detecting isolated keywords.

A.2 Fine-Tuning Procedure

The training set consists of 300 collective agreements manually annotated by a research assistant after full reading of each document. The annotation task is binary: does the agreement establish remote work provisions for the firm’s employees? Ambiguous cases were resolved through discussion and coded as negative, as they do not reflect a general organizational policy.

The fine-tuning was performed using Low-Rank Adaptation (LoRA; [Hu et al., 2022](#)), which freezes the pre-trained model weights and injects trainable low-rank matrices into each attention layer. The specific prompt used during training was: Does the content of this agreement allow us to affirm that remote work is being implemented in this company? The expected response takes the value Remote work or No remote work.

Hyperparameters: LoRA rank $r = 16$, scaling coefficient $\alpha = 32$, learning rate $\eta = 2 \times 10^{-4}$, batch size 4, 3 training epochs. Training was performed on a single A100 GPU and completed in approximately 6 hours.

A.3 Error Analysis

Residual errors on the 100-agreement test set (2 false negatives and 3 false positives) share a common pattern: they involve agreements that mention the possibility of remote work for restricted populations (e.g., pregnant workers, employees affected by severe weather, workers with disabilities) without establishing a general remote work framework. Given their marginal frequency in the full corpus, they are unlikely to materially affect our estimates.

A.4 Structured Extraction of Remote Work Modalities

Beyond the binary classification, the model extracts structured information about authorized remote work intensity using a JSON output format with the following schema:

```
{
  "relevant": true/false,
  "remote_work_days": {
    "quantity": "2",
    "unit": "week" | "month" | "year",
    "precision": "maximum" | "minimum" | "fixed" | "annual cap"
  },
}
```

```
"eligible_population": "All employees" | "Managers only" | ...
}
```

The `precision` field distinguishes agreements setting an exact number of days (fixed), a ceiling (maximum), a floor (minimum), or an annual quota. We harmonize all time units into days per week, assuming 47 working weeks per year. When an agreement specifies multiple regimes, we retain the modal regime. The intensity variable is available for 38 percent of agreements classified as establishing remote work; the remaining 62 percent do not explicitly state the number of authorized days.

A.5 Distribution of Authorized Remote Work Intensity

Figure [A1](#) displays the distribution of authorized days per week among the 38 percent of remote work agreements reporting this information (approximately 6,900 firms). The distribution is strongly concentrated at two days per week (6,426 firms), confirming that the median reflects a genuine mass point rather than a smooth distribution. The right tail is thin: only 488 and 570 firms authorize four and five days respectively, so the negative coefficients at high intensity in Table [9](#) are estimated on smaller subsamples and should be interpreted with appropriate caution.

Figure A1: Distribution of Authorized Remote Work Intensity



Notes: Number of firms by authorized remote work days per week, among the 38% of remote work agreements reporting intensity. Total: approximately 6,900 firms. Source: ACCO-DTD Légifrance, author's calculations.

Table A1: Callaway-Sant’Anna Estimates: Effect of Remote Work on TFP

Period	Estimate	Std. Error
<i>Panel A: Pre-treatment periods</i>		
$t - 5$	−0.007	(0.009)
$t - 4$	−0.005	(0.006)
$t - 3$	−0.008	(0.005)
$t - 2$	0.007	(0.005)
$t - 1$	0.003	(0.005)
<i>Panel B: Post-treatment periods</i>		
$t = 0$	0.010	(0.005)
$t + 1$	0.006	(0.007)
$t + 2$	0.020	(0.009)
$t + 3$	0.020	(0.011)
$t + 4$	0.029	(0.013)
$t + 5$	0.041	(0.021)
<i>Panel C: Aggregated ATTs</i>		
Overall ATT (simple aggregation)	0.016*	(0.007)
Overall ATT (dynamic aggregation)	0.021*	(0.008)
Estimator	Callaway-Sant’Anna (2021)	
Identification	Doubly Robust	
Comparison group	Not Yet Treated	
Panel	Balanced 2017–2024	
Observations	46,600	
Firms	5,825	

Notes: [Callaway and Sant’Anna \(2021\)](#) doubly robust estimator. Comparison group: not-yet-treated firms. Balanced panel, 2017–2024. TFP estimated via [Akerberg et al. \(2015\)](#) with wage bill as labor measure. * denotes that the simultaneous confidence band does not cover 0.

Table A2: Sun and Abraham Input Decomposition (Full Panel, For Reference)

	TFP (1)	Output (2)	Capital (3)	Labour (4)	Interm. (5)
$t - 5$	0.014* (0.010)	-0.043*** (0.013)	0.015 (0.020)	-0.059*** (0.013)	-0.103*** (0.019)
$t - 3$	0.003 (0.005)	-0.024*** (0.007)	0.014 (0.011)	-0.028*** (0.007)	-0.049*** (0.010)
$t - 2$	0.004 (0.004)	-0.007 (0.005)	0.008 (0.007)	-0.011** (0.005)	-0.028*** (0.007)
$t = 0$	0.007· (0.004)	0.023*** (0.005)	-0.012* (0.007)	0.017*** (0.004)	0.028*** (0.005)
$t + 2$	0.012· (0.007)	-0.003 (0.009)	-0.081*** (0.015)	-0.012 (0.009)	0.009 (0.010)
$t + 3$	0.023** (0.009)	-0.012 (0.011)	-0.131*** (0.020)	-0.030*** (0.011)	-0.005 (0.013)
$t + 5$	0.038* (0.017)	-0.055** (0.022)	-0.241*** (0.041)	-0.084*** (0.023)	-0.052* (0.027)
Firm FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
IPW	Yes	Yes	Yes	Yes	Yes
Obs.	105,876	105,876	105,876	105,876	105,876
R^2	0.879	0.952	0.947	0.942	0.939

Notes: [Sun and Abraham \(2021\)](#) estimator with IPW. Reference: $t - 1$. Full unbalanced panel, 14,551 firms. Pre-trends on output, labor, and intermediate inputs at $t - 5$ and $t - 3$ reflect sample composition at long lags. · $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table A3: Aggregate Effect of Remote Work: Robustness to Labor Input Measure

	TFP (ACF, headcount)	
	TWFE	Sun-Abraham
	(1)	(2)
Remote work	0.019*** (0.006)	0.015 (0.010)
Observations	105,886	105,886
Adj. R^2	0.927	—
Firm FE	Yes	Yes
Year FE	Yes	Yes

Notes: TFP estimated via [Akerberg et al. \(2015\)](#) with headcount as labor measure. Column (1): TWFE. Column (2): [Sun and Abraham \(2021\)](#), never-treated comparison group. Firm-clustered SE. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B Robustness Checks

B.1 Callaway-Sant’Anna Estimates: TFP

B.2 Sun and Abraham Input Decomposition

B.3 Alternative Labor Input Measure

B.4 Sector $\times$ Year Fixed Effects

B.5 Placebo Test

B.6 TFP Measure: Corrected vs. Uncorrected

B.7 Alternative TFP Specification: Calibrated Gross-Output Residual

Table [A7](#) reports event-study estimates from the Sun and Abraham (2021) estimator using an alternative TFP measure constructed as the residual of a three-input gross-output Cobb-

Table A4: Dynamic Effects: Sector $\times$ Year Fixed Effects

	Year FE (1)	Sector $\times$ Year FE (2)
$t - 5$	0.014 (0.010)	0.011 (0.010)
$t - 4$	0.002 (0.007)	-0.002 (0.008)
$t - 3$	0.003 (0.005)	0.001 (0.005)
$t - 2$	0.004 (0.004)	0.003 (0.004)
$t = 0$	0.007 (0.004)	0.007 (0.004)
$t + 1$	0.004 (0.005)	0.007 (0.005)
$t + 2$	0.012 (0.007)	0.017** (0.007)
$t + 3$	0.023** (0.009)	0.028*** (0.009)
$t + 4$	0.028* (0.012)	0.037*** (0.012)
$t + 5$	0.038* (0.017)	0.053*** (0.018)
Firm FE	Yes	Yes
Year FE	Yes	No
Sector $\times$ Year FE	No	Yes
Observations	105,876	105,810
R^2	0.879	0.898

Notes: [Sun and Abraham \(2021\)](#) estimator with IPW. TFP via [Akerberg et al. \(2015\)](#), wage bill as labor measure. Reference: $t - 1$. Sectors at two-digit NAF. Firm-clustered SE.
 * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Placebo Test: Fictitious Adoption Dates ($t_{\text{adopt}} - 3$)

	TFP (wage bill)
Placebo treatment	−0.009 (0.006)
Firm FE	Yes
Year FE	Yes
Observations	77,927
Adj. R^2	0.896

Notes: Sample restricted to pre-treatment periods. Fictitious adoption date set three years before actual adoption. TWFE with IPW. TFP via [Akerberg et al. \(2015\)](#), wage bill as labor measure. Firm-clustered SE.

Table A6: Robustness to TFP Correction for Intermediate Inputs

	Corrected TFP ($\tilde{\omega}$) (1)	Uncorrected TFP ($\hat{\omega}$) (2)
$t + 3$	0.023** (0.009)	0.023** (0.009)
$t + 4$	0.028* (0.012)	0.029* (0.012)
$t + 5$	0.038* (0.017)	0.040* (0.018)
Firm FE	Yes	Yes
Year FE	Yes	Yes
IPW	Yes	Yes
Observations	105,876	

Notes: [Sun and Abraham \(2021\)](#) estimator with IPW. Column (1): corrected TFP $\tilde{\omega}_{it} = \hat{\omega}_{it} - \hat{\alpha}_M m_{it}$. Column (2): raw `omega()` output. Pre-treatment coefficients insignificant in both specifications and omitted. Firm-clustered SE. * $p < 0.05$, ** $p < 0.01$.

Douglas production function:

$$\tilde{\omega}_{it} = \ln Y_{it} - \hat{\alpha}_{l,s} \ln L_{it} - \hat{\alpha}_{k,s} \ln K_{it} - \hat{\alpha}_{m,s} \ln M_{it} \quad (5)$$

where $\hat{\alpha}_{l,s}$, $\hat{\alpha}_{k,s}$, and $\hat{\alpha}_{m,s}$ are sector-specific input elasticities calibrated on the control group under the constant-returns constraint $\hat{\alpha}_{l,s} + \hat{\alpha}_{k,s} + \hat{\alpha}_{m,s} = 1$. Labor and capital elasticities are estimated by within-firm OLS on the never-treated and not-yet-treated comparison group, separately by sector (NA10 classification). The intermediate input elasticity is recovered as the residual share. This specification follows [Fiori et al. \(2025\)](#), who apply an analogous calibrated residual approach to French firm-level data from CIQ Pro.

Table A7: Event-Study Estimates: Alternative TFP (Calibrated Gross-Output Residual)

Event time	Estimate	Std. Error	<i>t</i> -value	<i>p</i> -value
<i>t</i> − 5	−0.002	0.009	−0.181	0.857
<i>t</i> − 4	−0.004	0.007	−0.687	0.492
<i>t</i> − 3	−0.003	0.004	−0.776	0.438
<i>t</i> − 2	−0.011	0.003	−3.574	0.000
<i>t</i> = 0	+0.005	0.003	+1.782	0.075
<i>t</i> + 1	+0.003	0.004	+0.855	0.392
<i>t</i> + 2	+0.011	0.005	+2.065	0.039
<i>t</i> + 3	+0.017	0.007	+2.450	0.014
<i>t</i> + 4	+0.017	0.008	+2.041	0.041
<i>t</i> + 5	+0.059	0.017	+3.419	0.001
<i>Aggregated ATT</i>				
ATT	+0.0075	0.0029	+2.586	0.010
Observations: 172,576		Firms: 13,057	Within <i>R</i> ² : 0.003	

Notes: Sun and Abraham (2021) estimator with firm and year fixed effects. TFP constructed as in equation (5) with sector-specific elasticities calibrated on the control group. Standard errors clustered at the firm level. Never-treated and not-yet-treated firms serve as the comparison group.

Table A8 reports the sensitivity of the aggregated ATT to perturbations of the calibrated elasticities. The ATT remains positive and significant at the 1% level across all four alternative calibrations, confirming that the main result is not driven by the specific elasticity

values chosen.

Table A8: Sensitivity of ATT to Calibrated Elasticity Values

Specification	ATT	Std. Error	p -value
Baseline	+0.0075	0.0029	0.010
$\hat{\alpha}_m + 0.05$	+0.0072	0.0028	0.010
$\hat{\alpha}_m - 0.05$	+0.0078	0.0030	0.010
$\hat{\alpha}_k + 0.05$	+0.0108	0.0031	0.000

Notes: Each row perturbs one elasticity by ± 5 percentage points relative to the baseline calibration, adjusting the labor elasticity $\hat{\alpha}_l$ to maintain $\sum_j \hat{\alpha}_j = 1$. Sun and Abraham (2021) estimator with firm and year fixed effects, clustered standard errors.

B.8 Permutation Test

B.9 Excluding the COVID-19 Adoption Cohort

B.10 Cohort-Specific ATTs and Pre-Pandemic Adopters

Table A9 reports cohort-specific ATTs from the Sun-Abraham estimator on the corrected panel. The never-treated group includes all firms with no remote work agreement throughout the observation window ($c_i = \infty$) and the 100 firms whose first agreement predates the panel start (2017 adopters), whose pre-treatment dynamics are entirely unobserved. The 2019 cohort yields the largest and most precisely estimated cohort-specific ATT (+5.0 pp, $p < 0.01$), reflecting its unique position as the only pre-pandemic cohort with sufficient post-adoption years to reach $t + 5$ for all treated firms. Cohorts adopting from 2020 onward show progressively smaller and less precisely estimated ATTs. This pattern is mechanically explained by the panel horizon: since the aggregate efficiency gain materializes only from $t + 3$ onward (Figure 1), cohorts adopting in 2022, 2023, 2024, and 2025 reach at most $t + 3$, $t + 2$, $t + 1$, and $t = 0$ respectively by the end of the panel in 2025, placing them entirely below the threshold at which the restructuring channel generates detectable gains. The COVID-19

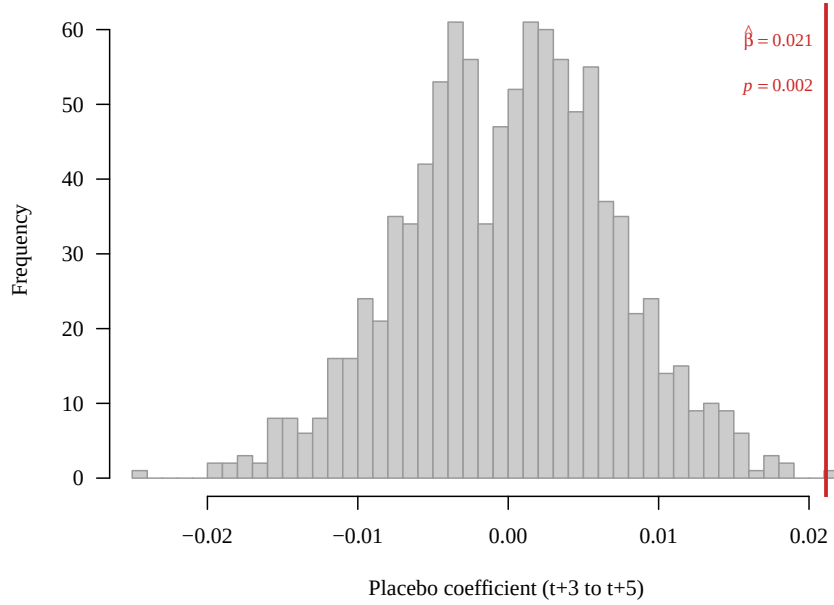


Figure A2: Permutation Test: Placebo Distribution vs. Actual Estimate

Notes: Distribution of 1,000 placebo coefficients from random reassignment of adoption dates. Treatment effect estimated at horizons $t + 3$ to $t + 5$ using TWFE with IPW. The vertical line indicates the actual estimate. Two-sided $p = 0.002$.

cohort (2020) shows a near-zero ATT (-0.6 pp), consistent with the absence of deliberate reorganization intent documented in Figure 5.

Table A10 further restricts the analysis to pre-pandemic adopters only (cohorts 2018 and 2019, $N = 1,482$ treated firms), using the 6,158 never-treated firms as the comparison group. These firms adopted for reasons entirely unrelated to the health crisis and have the longest post-adoption horizon in the panel. Pre-treatment coefficients are flat throughout the $[-4, -1]$ window, with a joint Wald test on $[-4, -2]$ yielding $F = 0.42$ ($p = 0.74$). The efficiency gain is monotonically increasing from $t = 0$ to $t + 5$, reaching $+7.0$ pp at $t + 5$ ($p < 0.01$), and is statistically significant from $t = 0$ onward. The larger magnitude relative to the full-sample estimate ($+3.8$ pp) is consistent with the panel truncation interpretation: early adopters are the only cohort for which the full dynamic trajectory of the restructuring mechanism is observable within the current panel window.

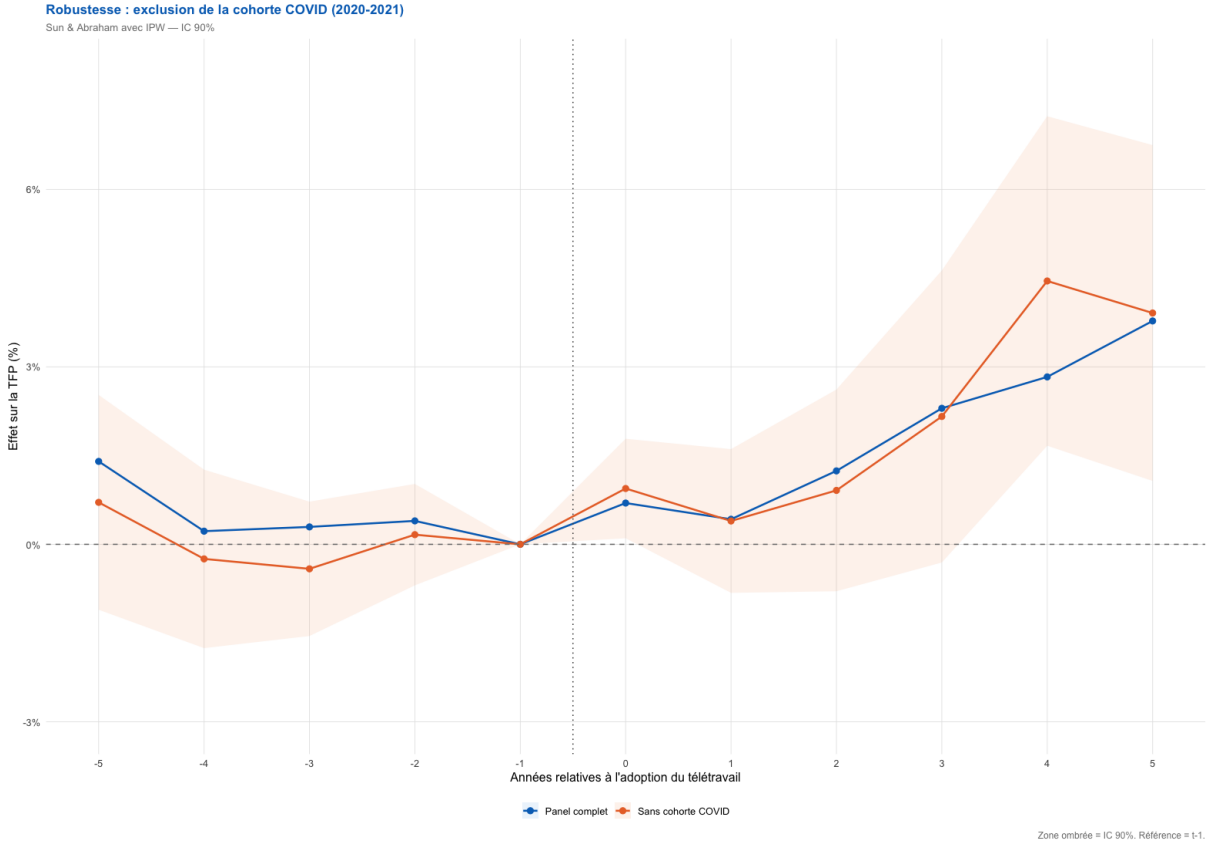


Figure A3: Robustness to Exclusion of the COVID-19 Adoption Cohort (2020–2021)

Notes: Sun and Abraham (2021) estimator with IPW. Blue line: baseline ($N = 105,876$). Orange line: excluding 2020–2021 adopters ($N = 78,593$). Shaded areas: 90% confidence intervals, firm-clustered SE. Reference period: $t - 1$.

Table A9: ATT by Adoption Cohort: Sun-Abraham Estimates

Cohort	ATT	Std. Error	p -value	N firms	Max horizon
2019	+0.0496	(0.0185)	0.007	830	$t + 6$
2020	−0.0063	(0.0077)	0.414	2,021	$t + 5$
2021	+0.0144	(0.0094)	0.125	1,872	$t + 4$
2022	+0.0044	(0.0084)	0.601	1,370	$t + 3$
2023	+0.0041	(0.0099)	0.680	989	$t + 2$
2024	+0.0277	(0.0159)	0.081	845	$t + 1$
2025	−0.0404	(0.1878)	0.830	297	$t = 0$
Aggregate ATT	+0.0081	(0.0037)	0.027	8,224	

Notes: Sun and Abraham (2021) estimator with IPW on the corrected panel ($c_i = \infty$ for never-adopters and pre-panel adopters). Cohort-specific ATTs are weighted averages of all post-adoption CATT estimates for the relevant cohort. Max horizon: maximum post-adoption year observed in the panel by end of 2025. Firm-clustered standard errors in parentheses.

Table A10: Event-Study Estimates: Pre-Pandemic Adopters Only (Cohorts 2018–2019)

Period	Estimate	Std. Error
<i>Panel A: Pre-treatment periods</i>		
$t - 4$	+0.008	(0.019)
$t - 3$	+0.003	(0.013)
$t - 2$	+0.005	(0.009)
<i>Panel B: Post-treatment periods</i>		
$t = 0$	+0.032*	(0.015)
$t + 1$	+0.035	(0.018)
$t + 2$	+0.048*	(0.021)
$t + 3$	+0.052*	(0.022)
$t + 4$	+0.073**	(0.025)
$t + 5$	+0.070**	(0.023)
<i>Panel C: Diagnostics</i>		
Joint pre-trend test $[-4, -2]$	$F = 0.42, p = 0.74$	
Estimator	Sun and Abraham (2021) with IPW	
Treated firms	1,482 (cohorts 2018–2019)	
Never-treated firms	6,158	
Observations	44,114	
Adj. R^2	0.908	

Notes: Sample restricted to pre-pandemic adopters (cohorts 2018 and 2019) and never-treated firms. Reference period: $t - 1$. Firm-clustered standard errors in parentheses. $\cdot p < 0.10$, * $p < 0.05$, ** $p < 0.01$.

B.11 Pooled TWFE by Authorized Intensity

Table A11 reports the pooled two-way fixed effects specification that estimates the effect of each authorized intensity on TFP using a single, common never-treated control group. This specification is the appropriate benchmark for comparing TFP levels across intensities, as it avoids the implicit control group contamination that arises when sub-sample regressions (as in Table 9) are used for cross-intensity comparisons. Each intensity indicator takes the value one for treated firm-years in which the authorized intensity falls within the corresponding bracket, and zero otherwise. Never-treated firms contribute to the control group for all intensity indicators simultaneously. The specification is estimated on the full panel of 105,886 observations with firm and year fixed effects and firm-clustered standard errors.

Table A11: Pooled TWFE Estimates of TFP by Authorized Remote Work Intensity

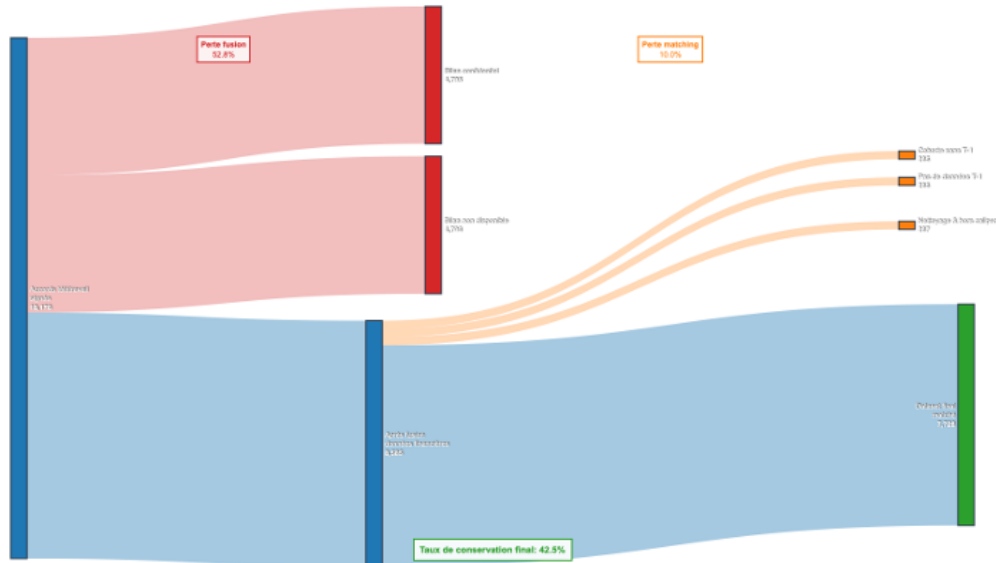
Authorized intensity	Estimate	Std. Error
1 day per week	+0.0260 [·]	(0.0154)
2 days per week	+0.0319**	(0.0118)
3 days per week	+0.0249	(0.0216)
4 days per week	−0.0026	(0.0281)
5 days per week	−0.0683**	(0.0536)
Observations	105,886	
Adj. R^2	0.9266	
Within R^2	0.00045	
Firm FE	Yes	
Year FE	Yes	

Notes: TWFE estimation with firm and year fixed effects on the full panel. The comparison group consists of never-treated firms common to all intensity categories. Each intensity indicator is mutually exclusive. Treated firms without intensity information are excluded; the remaining treated observations contribute to the relevant intensity indicator. Intensity categories: $[0.5, 1.5) = 1$ day, $[1.5, 2.5) = 2$ days, $[2.5, 3.5) = 3$ days, $[3.5, 4.5) = 4$ days, $\geq 4.5 = 5$ days. TFP estimated via [Akerberg et al. \(2015\)](#) with wage bill as labor measure. Firm-clustered standard errors in parentheses. [·] $p < 0.10$, * $p < 0.05$, ** $p < 0.01$. Note that standard TWFE is subject to the contamination bias identified by [Goodman-Bacon \(2021\)](#) in staggered adoption designs; these estimates should be interpreted as descriptive benchmarks for cross-intensity comparison rather than as cohort-robust causal estimates.

The results confirm a non-monotone TFP profile across intensities: the effect peaks at two authorized days per week (+3.2 pp, $p < 0.01$) and turns significantly negative at full remote work (−6.8 pp, $p < 0.05$). The one-day estimate (+2.6 pp, $p < 0.10$) is lower than the two-day estimate and does not individually achieve conventional significance thresholds, placing the productivity optimum unambiguously at two days. Three and four days show positive but statistically indistinguishable from zero effects, consistent with the interpretation that coordination costs begin to erode the real estate gain at these intensities without yet fully reversing it. Full remote work alone generates a statistically significant TFP loss. This non-monotone pattern is consistent with the two competing forces identified in Table 9: capital rationalization, which scales monotonically with intensity, and coordination costs on output, which materialize only at high intensity levels.

C Additional Tables and Figures

Figure A4: Sample Construction



Notes: Flow of observations from the initial sample to the final panel. Sources: ACCO-DTD Légifrance, INPI, author's calculations.

Table A12: Event-Study Estimates: Effect of Remote Work on Input Efficiency (TFP)

Period	Estimate	Std. Error
<i>Panel A: Pre-treatment periods</i>		
$t - 5$	0.014	(0.010)
$t - 4$	0.002	(0.007)
$t - 3$	0.003	(0.005)
$t - 2$	0.004	(0.004)
<i>Panel B: Post-treatment periods</i>		
$t = 0$	0.007·	(0.004)
$t + 1$	0.004	(0.005)
$t + 2$	0.012·	(0.007)
$t + 3$	0.023**	(0.009)
$t + 4$	0.028*	(0.012)
$t + 5$	0.038*	(0.017)
Firm FE	Yes	
Year FE	Yes	
IPW	Yes	
Observations	105,876	
Firms	14,551	
Adj. R^2	0.879	
Within R^2	0.004	

Notes: [Sun and Abraham \(2021\)](#) estimator with IPW. Reference period: $t - 1$. Firm-clustered SE. · $p < 0.10$, * $p < 0.05$, ** $p < 0.01$.

Table A13: Dynamic Effects on Production Function Inputs: Callaway-Sant’Anna Estimates

	TFP	Output	Capital	Labour	Interm.
	(1)	(2)	(3)	(4)	(5)
<i>Panel A: Pre-treatment periods</i>					
$t - 5$	−0.007 (0.009)	0.017 (0.018)	−0.006 (0.015)	0.025 (0.012)	0.036 (0.027)
$t - 3$	−0.008 (0.005)	−0.003 (0.010)	−0.013 (0.009)	0.005 (0.010)	0.005 (0.010)
$t - 2$	0.007 (0.005)	0.004 (0.006)	−0.010 (0.009)	−0.002 (0.004)	0.010 (0.009)
<i>Panel B: Post-treatment periods</i>					
$t = 0$	0.010 (0.005)	0.012 (0.006)	−0.016 (0.008)	0.003 (0.004)	0.009 (0.007)
$t + 2$	0.020 (0.009)	−0.013 (0.010)	−0.100* (0.017)	−0.029 (0.011)	−0.023 (0.013)
$t + 3$	0.020 (0.011)	−0.026 (0.015)	−0.139* (0.024)	−0.040* (0.012)	−0.040 (0.017)
$t + 5$	0.041 (0.021)	−0.027 (0.031)	−0.221* (0.054)	−0.059* (0.028)	−0.047 (0.039)
Estimator	Callaway-Sant’Anna (2021), Doubly Robust				
Comparison group	Not Yet Treated				
Panel	Balanced 2017–2024				
Observations	46,600				
Firms	5,825				

Notes: [Callaway and Sant’Anna \(2021\)](#) doubly robust estimator, not-yet-treated comparison group. Reference period: $t - 1$. Selected horizons within the $[-5; +5]$ window; full event-study in Figure 2. Simultaneous confidence bands, firm-clustered bootstrap. * confidence band does not cover 0.

Table A14: Effect of Remote Work on Input Efficiency (TFP) by Sector and Firm Size

	Manuf.	Constr.	Prof. serv.	ICT	Other serv.	Trade	Public	Finance	Real est.
<i>Panel A: All firms</i>									
Remote work	-0.024*** (0.009)	-0.023 (0.021)	0.054*** (0.014)	0.094*** (0.023)	-0.105*** (0.038)	-0.003 (0.013)	0.012 (0.019)	0.141*** (0.034)	0.098 (0.075)
<i>Panel B: SMEs</i>									
Remote work	-0.032 (0.023)	-0.093* (0.054)	-0.032 (0.022)	-0.002 (0.029)	0.012 (0.066)	-0.032* (0.016)	0.008 (0.040)	0.105 (0.066)	0.127 (0.077)
N firms	1,207	326	1,139	531	107	1,197	239	158	50
<i>Panel C: Mid-sized firms</i>									
Remote work	0.021 (0.018)	0.064 (0.061)	0.006 (0.023)	-0.030 (0.050)	-0.011 (0.099)	0.017 (0.012)	-0.050 (0.046)	0.042 (0.052)	-0.054 (0.102)
N firms	1,806	241	1,235	499	65	2,183	271	227	37
<i>Panel D: Large firms</i>									
Remote work	0.025 (0.032)	-0.019 (0.025)	0.062* (0.034)	-0.006 (0.061)	0.319 (0.315)	0.076** (0.031)	0.022 (0.029)	0.055 (0.177)	0.167 (0.206)
N firms	880	502	647	212	49	626	306	87	28
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Dependent variable: TFP estimated via [Akerberg et al. \(2015\)](#) with headcount as labor measure. Each column is a separate regression by sector (NA10 classification). Firm-clustered standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D Full Callaway-Sant’Anna Results: Labour Channel Decomposition

Tables A15 and A16 report the complete event-study and aggregate ATT estimates from the Callaway-Sant’Anna doubly robust estimator for headcount and the log wage bill, respectively. Table A17 reports the corresponding estimates for the social contribution ratio. All estimates use the not-yet-treated comparison group on the balanced panel of 5,825 firms over 2017–2024.

Table A15: Callaway-Sant’Anna Estimates: Effect of Remote Work on Headcount

Event time	Estimate	Std. Error	[95% Simult. Conf. Band]
<i>Panel A: Pre-treatment periods</i>			
$t - 5$	5,299.8	(74,052.1)	[−183,585.3 ; 194,184.8]
$t - 4$	−171,803.7	(58,816.6)	[−321,827.5 ; −21,779.8]*
$t - 3$	73,338.0	(43,576.0)	[−37,811.6 ; 184,487.6]
$t - 2$	−55,177.5	(34,912.8)	[−144,229.6 ; 33,874.7]
$t - 1$	62,511.5	(27,472.3)	[−7,562.3 ; 132,585.4]
<i>Panel B: Post-treatment periods</i>			
$t = 0$	9,562.6	(21,854.4)	[−46,181.5 ; 65,306.6]
$t + 1$	−19,986.8	(33,240.0)	[−104,772.3 ; 64,798.7]
$t + 2$	−55,147.7	(39,577.9)	[−156,099.2 ; 45,803.8]
$t + 3$	−100,783.3	(52,595.3)	[−234,938.3 ; 33,371.8]
$t + 4$	−189,369.2	(79,622.6)	[−392,462.8 ; 13,724.5]
$t + 5$	−140,773.6	(181,081.9)	[−602,660.1 ; 321,112.9]
<i>Panel C: Aggregated ATTs</i>			
Overall ATT (simple)	−56,355.1	(38,222.1)	[−131,269.0 ; 18,558.8]
Overall ATT (dynamic)	−82,749.7	(60,826.4)	[−201,967.1 ; 36,467.8]
Estimator	Callaway-Sant’Anna (2021), Doubly Robust		
Comparison	Not Yet Treated		
Anticipation	0 periods		

Notes: * denotes that the simultaneous confidence band does not cover 0. The pre-treatment spike at $t - 4$ reflects small cohort composition at long lags and should be interpreted with caution. The absence of significant post-treatment effects reflects limited statistical power due to the elevated share of missing headcount values in the liasse fiscale filings rather than a structural zero effect.

Table A16: Callaway-Sant'Anna Estimates: Effect of Remote Work on Log Wage Bill

Event time	Estimate	Std. Error	[95% Simult. Conf. Band]
<i>Panel A: Pre-treatment periods</i>			
$t - 5$	+0.025	(0.011)	$[-0.006 ; +0.055]$
$t - 4$	+0.022	(0.008)	$[+0.000 ; +0.043]^*$
$t - 3$	+0.005	(0.011)	$[-0.024 ; +0.035]$
$t - 2$	-0.002	(0.004)	$[-0.013 ; +0.009]$
$t - 1$	+0.005	(0.006)	$[-0.010 ; +0.020]$
<i>Panel B: Post-treatment periods</i>			
$t = 0$	+0.003	(0.004)	$[-0.008 ; +0.015]$
$t + 1$	-0.006	(0.006)	$[-0.023 ; +0.011]$
$t + 2$	-0.029	(0.011)	$[-0.059 ; +0.001]$
$t + 3$	-0.040	(0.012)	$[-0.072 ; -0.008]^*$
$t + 4$	-0.055	(0.017)	$[-0.101 ; -0.010]^*$
$t + 5$	-0.059	(0.027)	$[-0.133 ; +0.014]$
<i>Panel C: Aggregated ATTs</i>			
Overall ATT (simple)	-0.021	(0.008)	$[-0.038 ; -0.005]^*$
Overall ATT (dynamic)	-0.031	(0.011)	$[-0.053 ; -0.009]^*$
Estimator	Callaway-Sant'Anna (2021), Doubly Robust		
Comparison	Not Yet Treated		
Anticipation	0 periods		

Notes: * denotes that the simultaneous confidence band does not cover 0. Dependent variable: log wage bill from the liasse fiscale (INPI). Firm-clustered bootstrap standard errors.

Table A17: Callaway-Sant’Anna Estimates: Effect of Remote Work on Social Contributions Ratio

Event time	Estimate	Std. Error	[95% Simult. Conf. Band]
<i>Panel A: Pre-treatment periods</i>			
$t - 5$	+0.0003	(0.0008)	[−0.0017 ; +0.0024]
$t - 4$	+0.0008	(0.0008)	[−0.0015 ; +0.0031]
$t - 3$	−0.0008	(0.0010)	[−0.0034 ; +0.0019]
$t - 2$	−0.0007	(0.0006)	[−0.0023 ; +0.0009]
$t - 1$	+0.0007	(0.0005)	[−0.0007 ; +0.0021]
<i>Panel B: Post-treatment periods</i>			
$t = 0$	−0.0004	(0.0005)	[−0.0019 ; +0.0010]
$t + 1$	−0.0001	(0.0007)	[−0.0021 ; +0.0018]
$t + 2$	+0.0002	(0.0011)	[−0.0027 ; +0.0031]
$t + 3$	−0.0002	(0.0013)	[−0.0038 ; +0.0033]
$t + 4$	+0.0000	(0.0020)	[−0.0056 ; +0.0056]
$t + 5$	+0.0012	(0.0020)	[−0.0043 ; +0.0067]
<i>Panel C: Aggregated ATTs</i>			
Overall ATT (simple)	−0.0001	(0.0009)	[−0.0019 ; +0.0017]
Overall ATT (dynamic)	+0.0001	(0.0010)	[−0.0019 ; +0.0021]
Estimator	Callaway-Sant’Anna (2021), Doubly Robust		
Comparison	Not Yet Treated		
Anticipation	0 periods		

Notes: Dependent variable: social contributions divided by the wage bill (ratio), from the liasse fiscale (INPI). No coefficient is statistically distinguishable from zero at any horizon, confirming the absence of any compositional shift toward lower-contribution contract types among treated firms. Firm-clustered bootstrap standard errors.

E Attrition Analysis

Table A18 compares firms retained in the final panel with those lost at the merge stage. The treated retention rate is 10.8 percentage points below that of controls (50.5 percent versus 61.3 percent), but the treated indicator is statistically insignificant in a logistic regression of the probability of attrition on all observables (coefficient -0.048 , $p = 0.585$; McFadden pseudo- $R^2 = 0.011$), indicating that the gap reflects the greater propensity of larger treated firms to file consolidated rather than individual accounts rather than differential selection on efficiency.

Table A18: Attrition Analysis: Retained vs. Lost Firms

Variable	Retained			Lost			SMD	p
	Mean	SD	N	Mean	SD	N		
Number of agreements	9.856	10.241	16,866	11.419	30.538	13,848	-0.069	0.000^{***}
Year of first agreement	2019.244	1.917	16,866	2019.282	2.085	13,848	-0.019	0.101
Log production (mean)	16.584	1.576	15,303	17.051	1.808	612	-0.276	0.000^{***}
Log capital (mean)	13.601	2.268	15,589	14.039	2.597	631	-0.180	0.000^{***}
Log wage bill (mean)	14.954	1.274	15,276	15.291	1.472	604	-0.245	0.000^{***}
Share treated	0.544		16,866	0.650		13,848	-0.217	0.000^{***}

Notes: Comparison of firms retained in the final panel ($N = 16,866$) against firms lost at the ACCO-INPI merge stage ($N = 13,848$). Financial variables are pre-adoption averages for treated firms and full-period averages for controls. SMD: standardized mean difference. p -values from two-sided Welch t -tests (continuous) and chi-squared tests (proportions). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A19 reports sensitivity bounds under alternative assumptions about the TFP of lost firms.

F IPW Covariate Balance

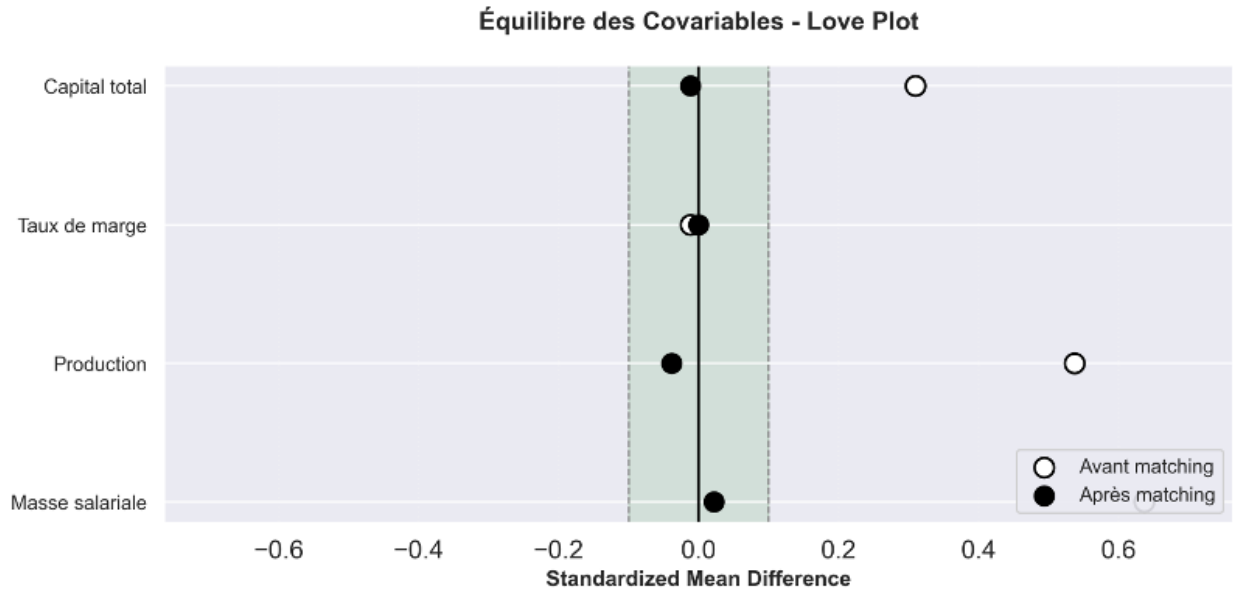
Figure A5 displays the standardized mean differences (SMD) for each covariate used in the propensity score estimation, before and after IPW reweighting. After reweighting, all SMDs fall within the conventional $[-0.1, 0.1]$ balance threshold, confirming that the IPW procedure successfully adjusts for observable selection on size, wage bill, capital intensity, and profitability.

Table A19: Sensitivity Bounds: ATT Under Alternative Attrition Assumptions

Assumed TFP gap	Implied bias	Adjusted ATT	Positive?	>1.5 pp?
−20%	−0.022	0.060	Yes	Yes
−10%	−0.011	0.049	Yes	Yes
−5%	−0.005	0.043	Yes	Yes
0%	0.000	0.038	Yes	Yes
+5%	+0.005	0.033	Yes	Yes
+10%	+0.011	0.027	Yes	Yes
+15%	+0.016	0.022	Yes	Yes
+20%	+0.022	0.016	Yes	Yes

Notes: Baseline ATT = 0.038. Implied bias = $0.108 \times \delta$, where 0.108 is the retention rate differential. Adjusted ATT = baseline ATT minus implied bias.

Figure A5: Covariate Balance Before and After IPW Reweighting

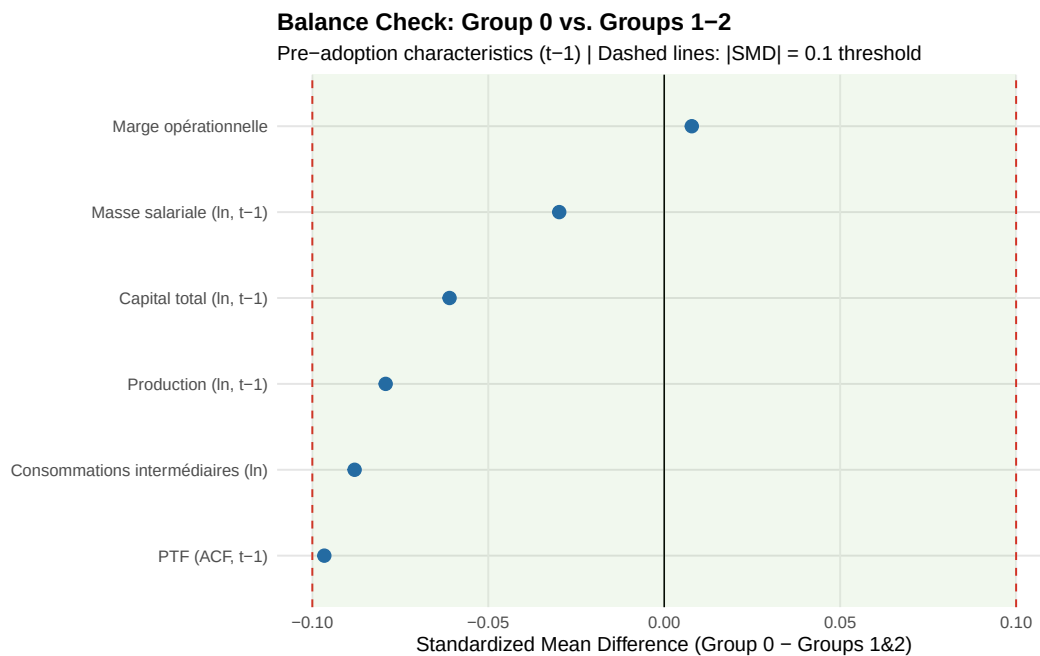


Notes: Open circles: SMD before reweighting. Filled circles: SMD after IPW reweighting. The green band delimits the $[-0.1, 0.1]$ conventional balance threshold. Covariates estimated via logit separately by adoption cohort on lagged $(t - 1)$ values. Source: author's calculations.

G Covariate Balance Across Residual Lease Maturity Groups

Figure A6 displays the standardized mean differences between firms with imminent lease renewal at adoption (Group 0, residual term ≤ 1 year) and firms with longer residual terms (Groups 1–2), measured on pre-adoption characteristics ($t - 1$). All standardized mean differences fall within the $[-0.10, 0.10]$ conventional balance threshold (Imbens and Rubin, 2015), confirming that residual lease maturity at adoption is conditionally orthogonal to firm characteristics. The uniform distribution of firms across the three maturity categories ($\chi^2 = 1.48$, $df = 2$, $p = 0.476$) and the covariate balance jointly support the quasi-experimental interpretation of the within-treated contrast in Figure 4.

Figure A6: Covariate Balance Across Residual Lease Maturity Groups



Notes: Standardized mean differences between firms with residual lease term ≤ 1 year at adoption (Group 0) and firms with residual lease term > 1 year (Groups 1–2), measured on pre-adoption characteristics ($t - 1$). Dashed lines: $|SMD| = 0.10$ conventional balance threshold. Green band: balance zone. Sample restricted to establishments registered after 2011 in SIRENE. Source: author's calculations.

H Full Callaway-Sant’Anna Results by Firm Size

Tables A20, A21, and A22 report the complete set of Callaway-Sant’Anna aggregated ATTs for each firm size category across all production function inputs and labour outcomes. These estimates underlie the summary Table 10 in Section 6. All estimates use the never-treated comparison group on a balanced panel of cylindrised firms over 2017–2024.

Table A20: Callaway-Sant’Anna Estimates by Outcome: SMEs

Outcome	ATT	Std. Error	p-value	Firms
TFP (ACF)	−0.0076	(0.0113)	0.499	1,762
Production (ln)	−0.0245	(0.0152)	0.108	1,762
Capital (ln)	−0.0737	(0.0308)	0.017	1,762
Wage bill (ln)	−0.0140	(0.0130)	0.283	1,762
Intermediate inputs (ln)	−0.0222	(0.0193)	0.250	1,762
Effectif	−4,841	(30,775)	0.875	1,762
Wage bill / Effectif (ln)	−0.0311	(0.0584)	0.594	1,762
Social contrib. / Wage bill	−0.0026	(0.0021)	0.219	1,762
Estimator	Callaway-Sant’Anna (2021), Doubly Robust			
Comparison	Never Treated			
Panel	Balanced 2017–2024			

Notes: Simple aggregated ATT. SME defined as fewer than 250 employees. Firm-clustered bootstrap standard errors.

Table A21: Callaway-Sant’Anna Estimates by Outcome: Mid-Sized Firms (ETI)

Outcome	ATT	Std. Error	p-value	Firms
TFP (ACF)	+0.0043	(0.0100)	0.665	2,598
Production (ln)	−0.0242	(0.0138)	0.080	2,598
Capital (ln)	−0.0731	(0.0236)	0.002	2,598
Wage bill (ln)	−0.0265	(0.0119)	0.026	2,598
Intermediate inputs (ln)	−0.0424	(0.0166)	0.010	2,598
Effectif	−58,438	(68,072)	0.391	2,598
Wage bill / Effectif (ln)	−0.0129	(0.0570)	0.820	2,598
Social contrib. / Wage bill	+0.0002	(0.0009)	0.819	2,598
Estimator	Callaway-Sant’Anna (2021), Doubly Robust			
Comparison	Never Treated			
Panel	Balanced 2017–2024			

Notes: Simple aggregated ATT. ETI defined as 250–4,999 employees. Firm-clustered bootstrap standard errors. The stable social contribution ratio rules out any compositional shift toward lower-contribution contracts, confirming that the wage bill decline reflects headcount adjustment rather than wage compression.

Table A22: Callaway-Sant’Anna Estimates by Outcome: Large Firms (GE)

Outcome	ATT	Std. Error	p-value	Firms
TFP (ACF)	+0.0519	(0.0142)	0.000	1,421
Production (ln)	+0.0391	(0.0232)	0.092	1,421
Capital (ln)	−0.1312	(0.0320)	0.000	1,421
Wage bill (ln)	−0.0062	(0.0191)	0.744	1,421
Intermediate inputs (ln)	+0.0395	(0.0302)	0.191	1,421
Effectif	−173,171	(128,725)	0.179	1,421
Wage bill / Effectif (ln)	+0.0907	(0.0630)	0.150	1,421
Social contrib. / Wage bill	+0.0020	(0.0017)	0.225	1,421
Estimator	Callaway-Sant’Anna (2021), Doubly Robust			
Comparison	Never Treated			
Panel	Balanced 2017–2024			

Notes: Simple aggregated ATT. GE defined as 5,000 or more employees. Firm-clustered bootstrap standard errors. The stable wage bill and rising wage-per-worker ratio for large firms is consistent with a skill-upgrading effect accompanying real estate rationalization, a margin not detected for smaller size categories.